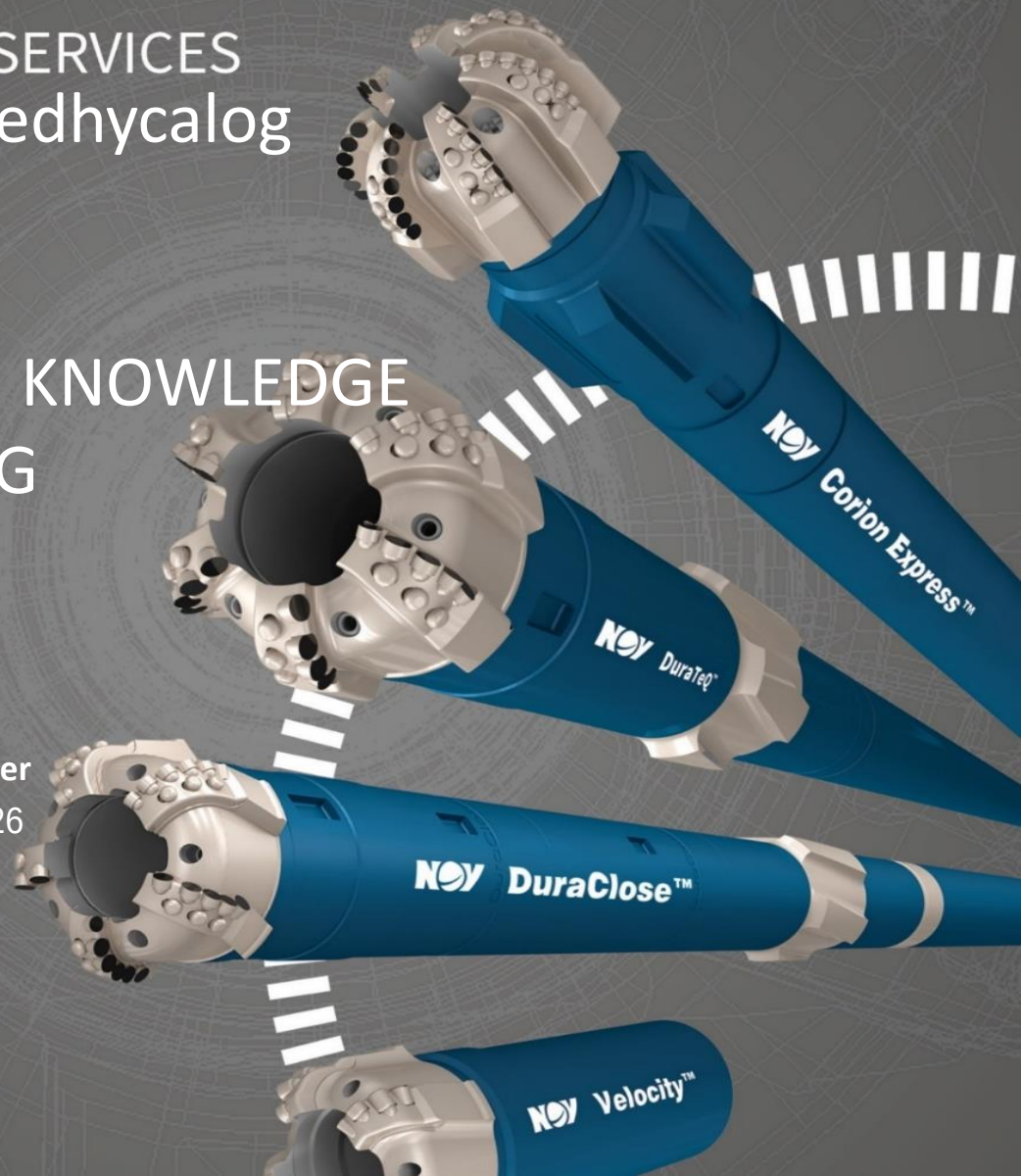


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CORING SERVICES

NOV Reedhycalog

CORING KNOWLEDGE SHARING



JDS - ASA Tower

1 Oktober 2026

By:

Benny Wales

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INTRODUCTION - ALL ATTENDESS

SAFETY MOMENT

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SAFETY SHARING
KESELAMATAN BERKENDARA

PT National Oilwell Varco

Safety Driving Campaign_Second Chance



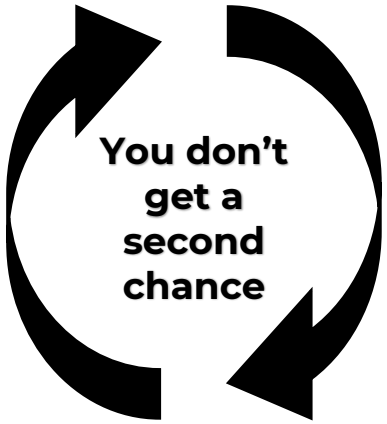
Safety Driving Campaign_Second Chance

Driving



Follow safe driving rules

1. I always wear a seatbelt
2. I do not exceed the speed limit, and reduce my speed for road conditions
3. I do not use distracting devices while driving
4. I am fit, rested, and fully alert while driving
5. I follow journey management requirements





Health, Safety, Security and Environment

Energy Products and Services

SAFETY SHARING

(699) SAFETY VIDEOS Sharing

Health, Safety, Security and Environment

Energy Products and Services

Coring Target Formations
Oil and Gas

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Reservoir Rocks

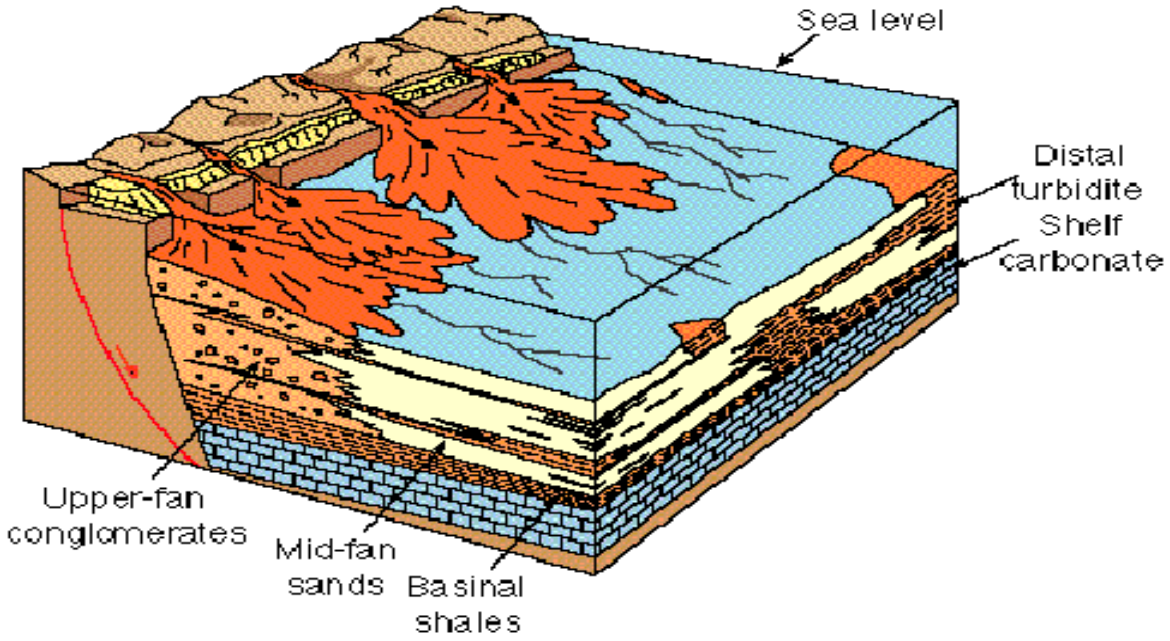
- Porous Sedimentary rocks.
- Pores hold oil and gas and also allow movement.
- Permeability is how easily the fluid flows.



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Sedimentary rocks

are types of rock that are formed by the deposition and subsequent cementation of that material at the Earth's surface and within bodies of water. Sedimentation is the collective name for processes that cause mineral and/or organic particles (detritus) to settle in place. The particles that form a sedimentary rock by accumulating are called sediment. Before being deposited, the sediment was formed by weathering and erosion from the source area, and then transported to the place of deposition by water, wind, ice, mass movement or glaciers, which are called agents of denudation. Sedimentation may also occur as minerals precipitate from water solution or shells of aquatic creatures settle out of suspension.

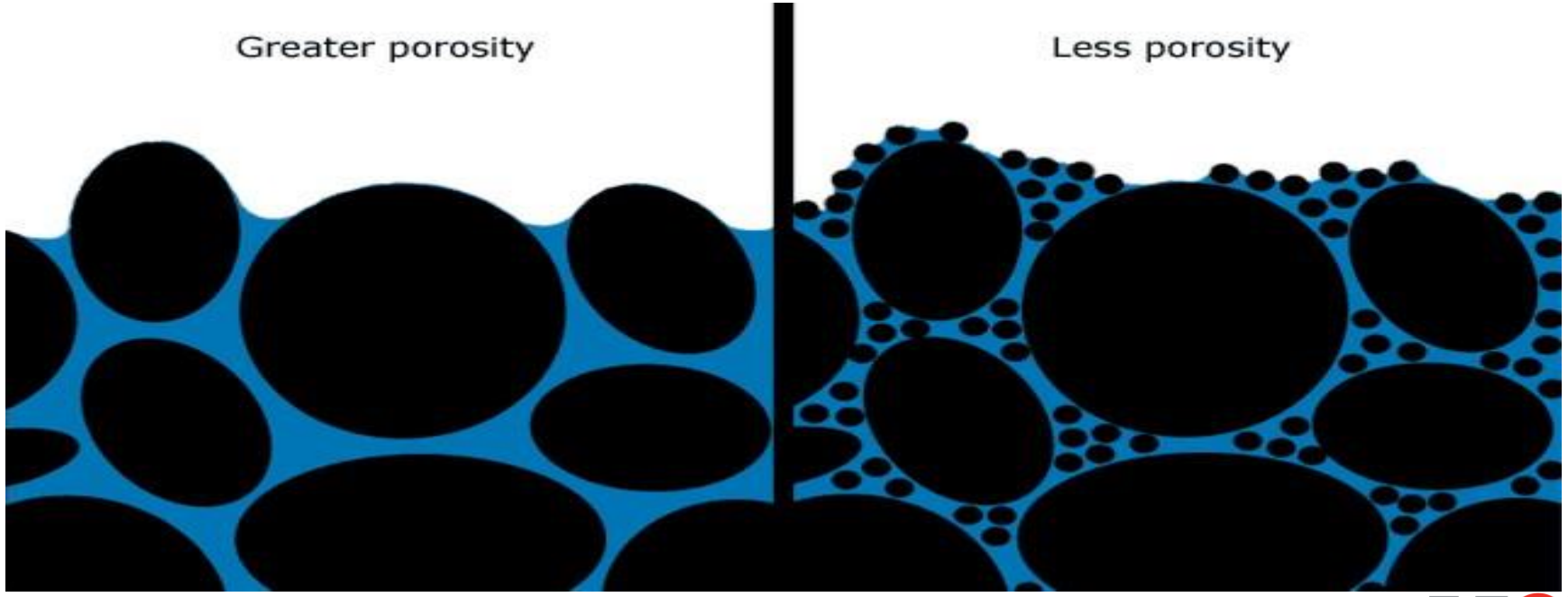


- Clastic - sandstone, siltstone, shale
- Non Clastic limestone, dolomite, salt, anhydrite etc.

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Porosity

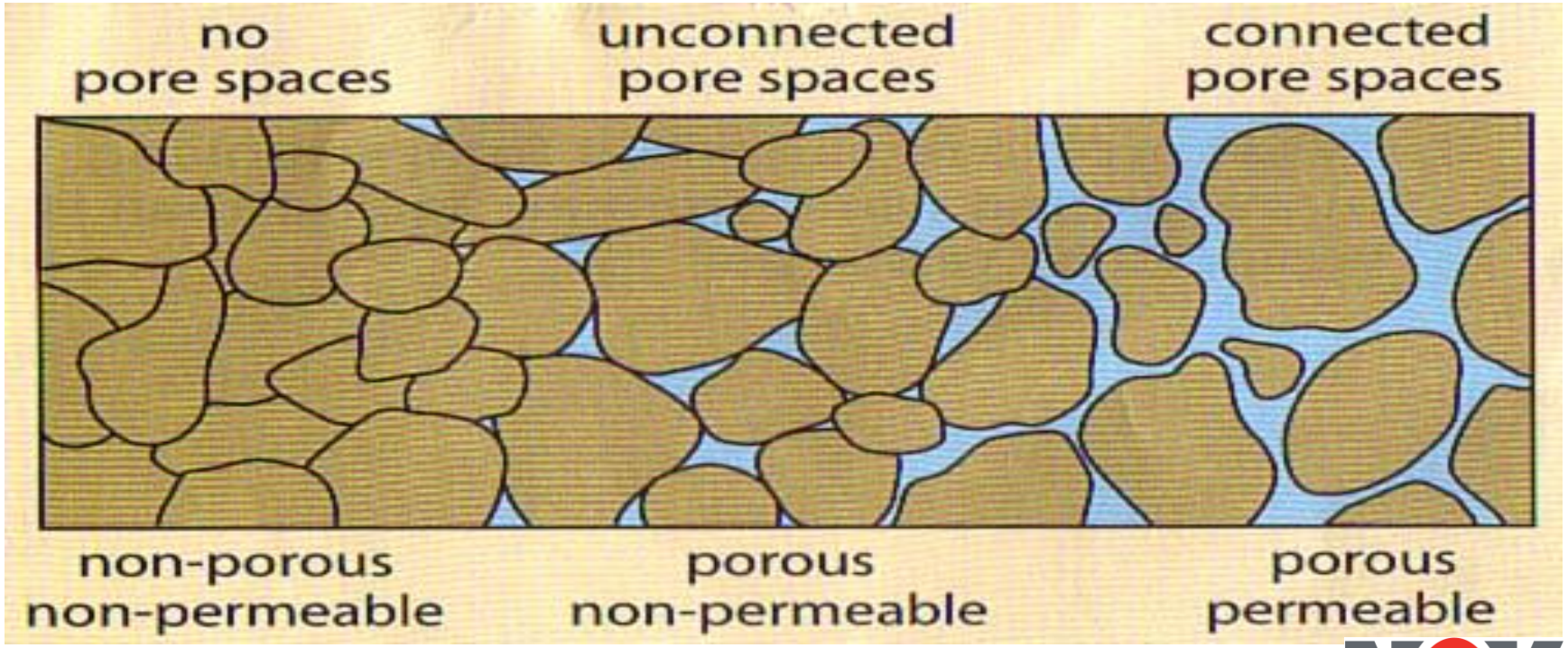
- Porosity is the percentage of the total volume of rock not occupied by mineral grains



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Permeability

- Permeability is the ability of a formation to allow the passage of pore fluid through the pore spaces.



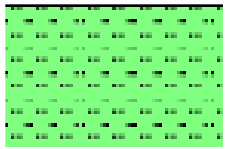
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Sandstone

- Often excellent reservoir rocks and PDC application
- Abrasiveness is related to cement type and grain shape and size.
- Stronger cements increase abrasively





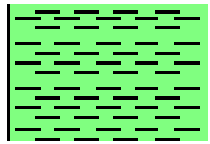


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Claystone Mudstone and Shale

- Very fine grained.
- Fails easily under shearing mechanism.
- Best drilled by PDC bits.
- Has the potential to ball bits in Water based mud





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Shale Facts

- 80% of formations drilled in oil and gas industry are shale's
- Shell estimates that 50% of lost drilling time is shale related either due to chemical or mechanical instability
- High Organic Content
- Excellent source rock
- Shears easily

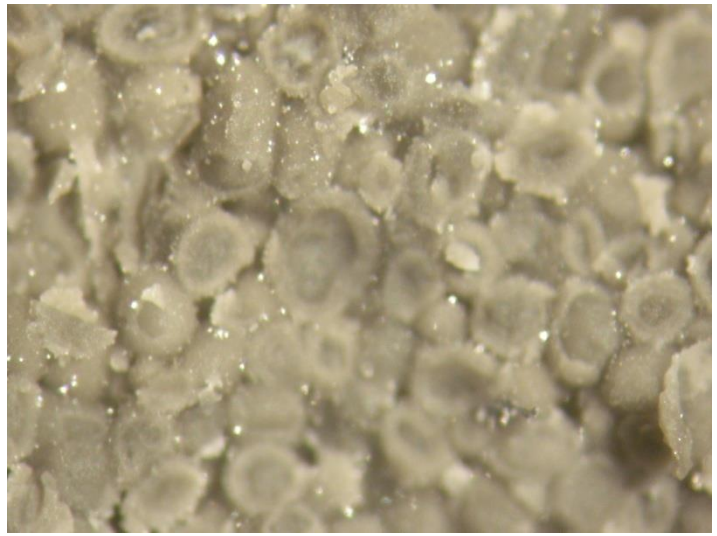




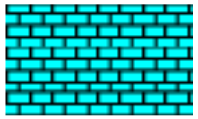
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Limestone CaCo3

- Limestone can often be fractured or cavernous and can cause lost circulation
- Can be soft to very hard and crystalline







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Siltstone

- A fine grained clastic formation grading between clay and a true sandstone
- Is distinguished from clay by “gritty” texture
- Can sometimes be hard and siliceous (silicon dioxide)



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Coal

- Fossilised plant remains
- High carbon content
- Gas source rock in some areas



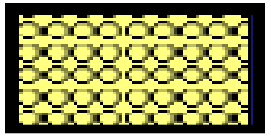
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Conglomerate / Breccia

- Is the largest grained clastic formation
- Grains can be made of numerous different igneous and metamorphic rock types
- Generally should be avoided with PDC
- Recently cored with PDC new technology







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Plutonic (Granite)

- Deep, slow-cooled crystalline rocks (e.g. Granite, Diorite, Gabbro)
- Can be effectively drilled with Impregnated Diamond Bit.
- Oil is occasionally produced from Fractured Granite and Granite Wash Reservoirs.
- Recently cored with PDC new technology



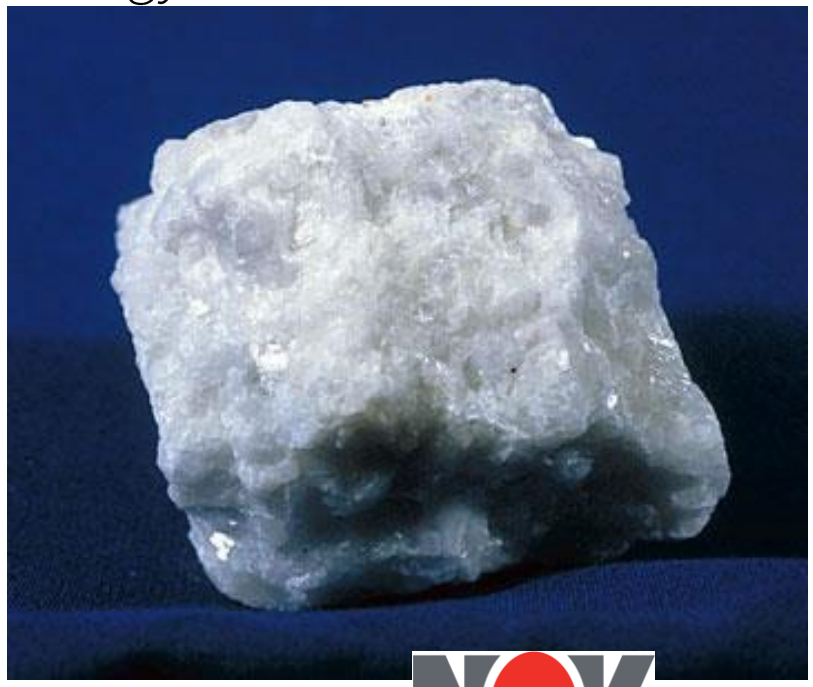
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Metamorphics Rock

- Altered rocks due to temperature and pressure
- Slate, Gneiss, Schist, Marble
- Not commonly drilled or cored in oilfields
- But more recent years cored with new technology







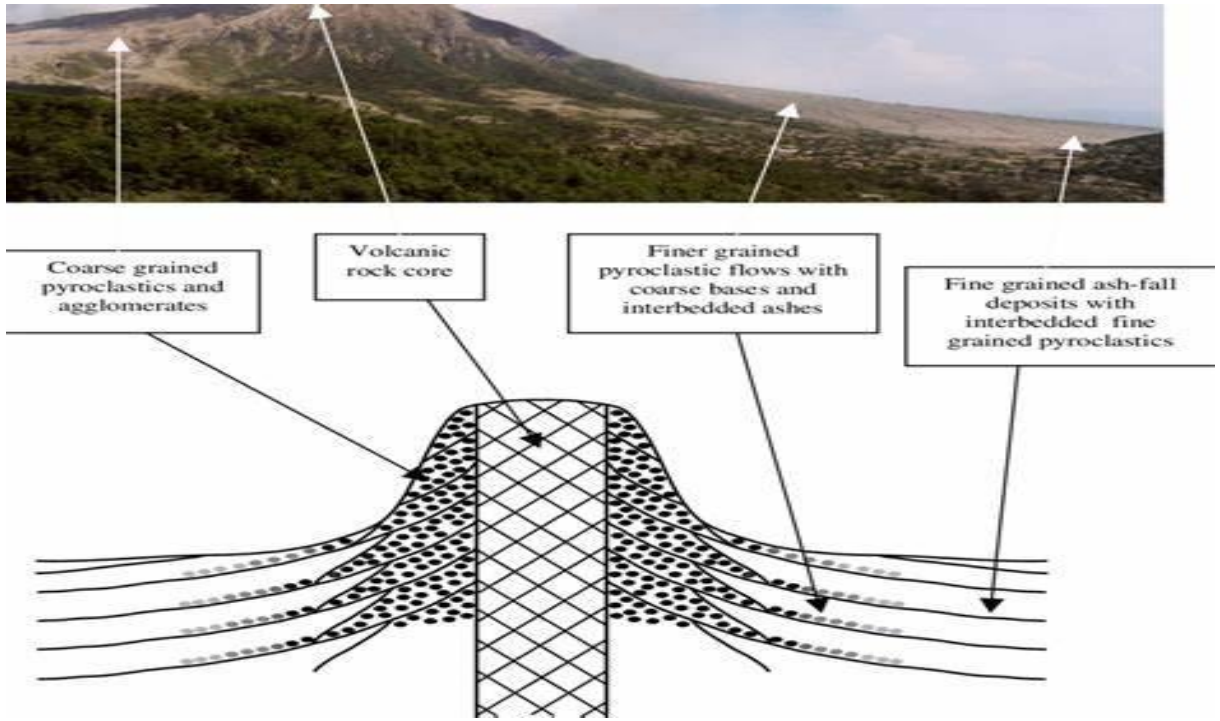
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Coring Target Formations Geothermal

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Andesite Lava

Andesite is a type of lava intermediate between rhyolite, typically associated with explosive volcanism, and basalt, which flows like honey along the earth's surface







Drilling and
on

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Volcanic - Pumice / Tuff

- Molten rock expelled at high temperature
- Quick solidification traps gasses
- Also classified as sedimentary rock



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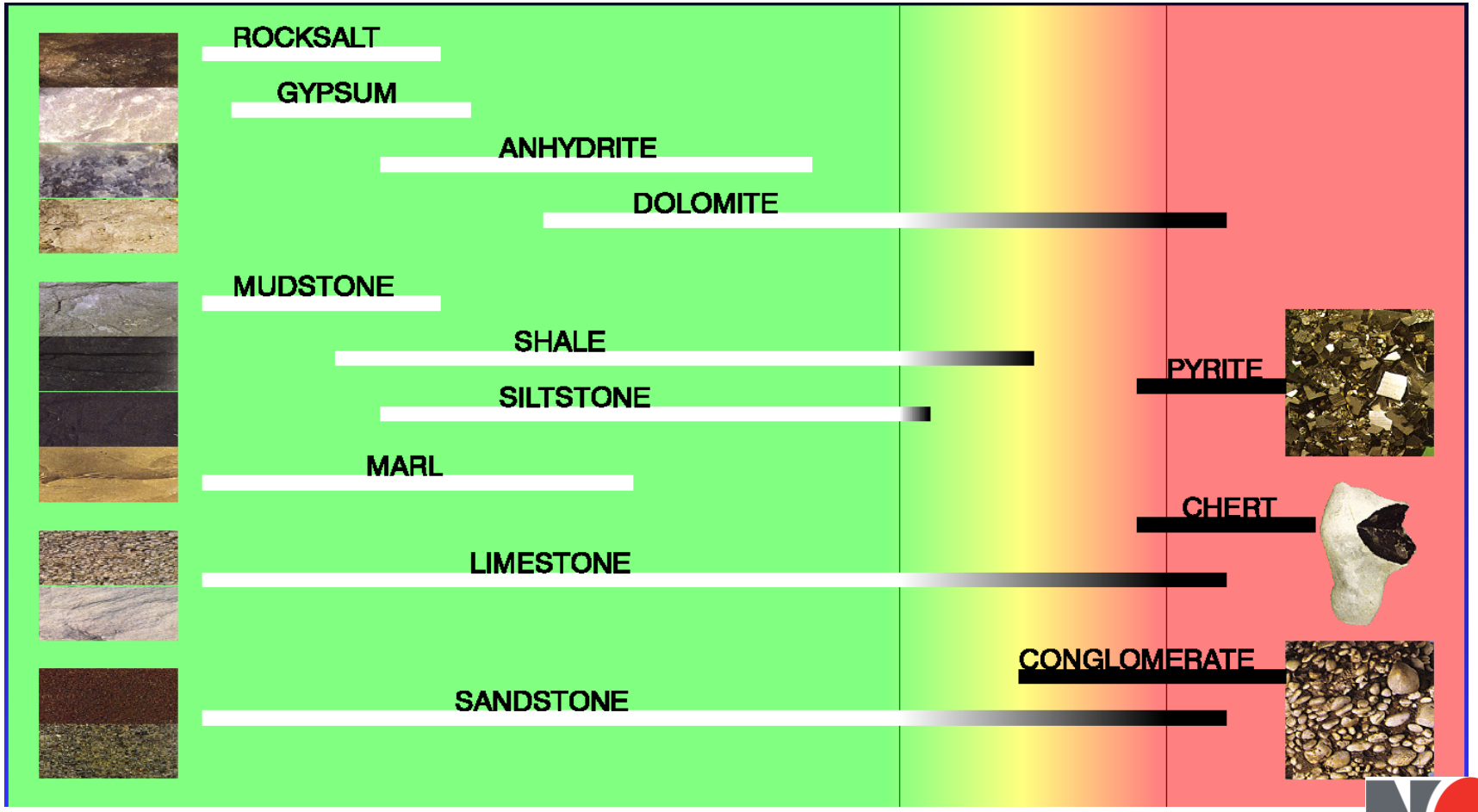
Formation Drillability

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Drillability of Sedimentary Rocks

PDC Drillable

Not PDC Drillable

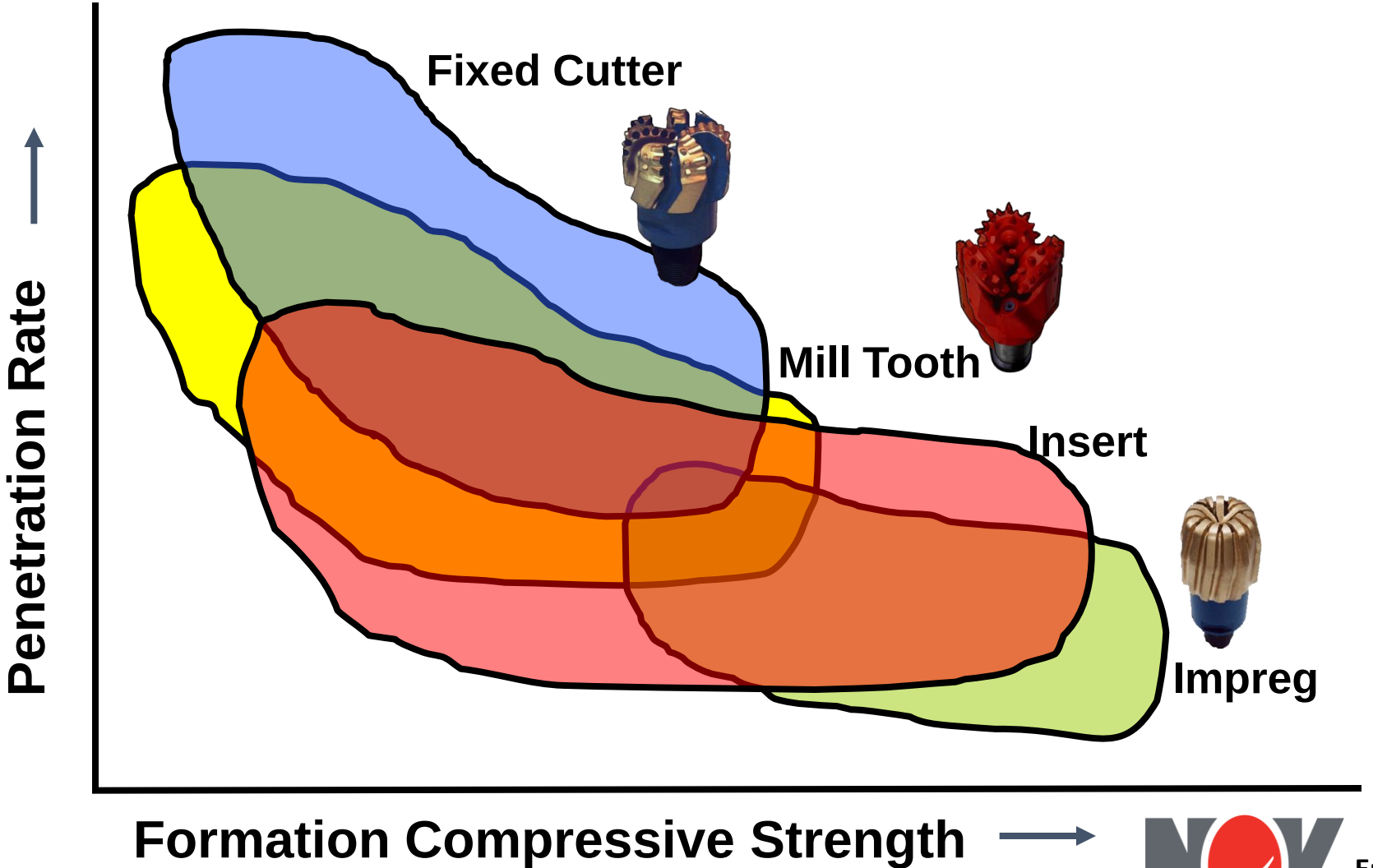


Recently Cored with PDC New Technology

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Bit Application Spectrum



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Coring Basic - Introduction

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Learning Objectives

At the end of this module you should be able to

- Define coring
- State the key events in the development of coring
- List key reservoir properties obtainable from core analysis
- Summarise the elements of the coring process
- Itemise the principal features of a coring assembly
- Note the more important problems that can occur while coring
- List the different kinds of coring and their purpose

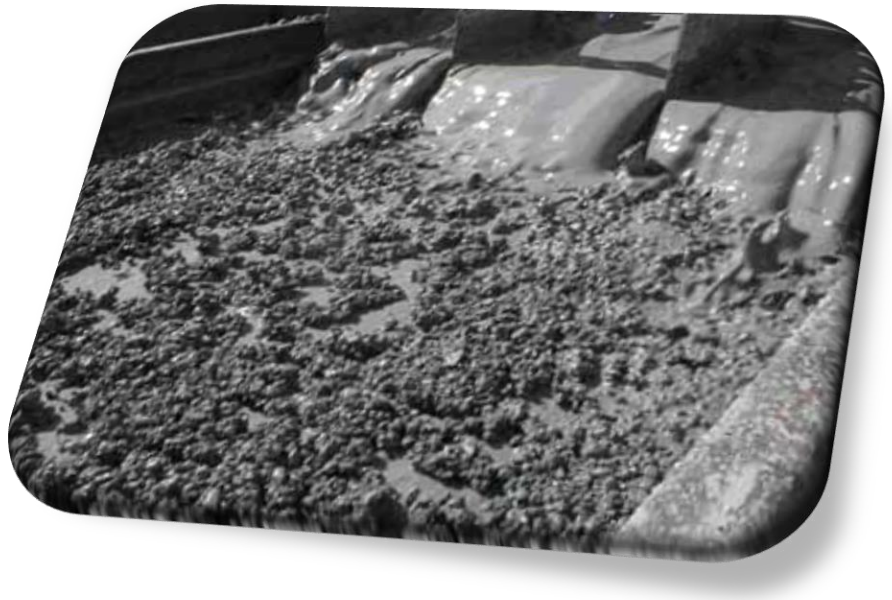
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What is coring?

Shale Shaker Video - Cuttings

Cuttings are :

- Small
- Invaded
- Mechanically damaged
- Mixed by depth



Coring :

is a process in which a continuous cylinder of rock is created by drilling an outer ring of rock in the borehole leaving behind the cylinder that is then retained within a tubular barrel and retrieved to surface.



"Size and Condition of the rock retrieved differentiate"

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What Is a core?

- Core is the Sample / Reservoir Rock itself







BAD Example
Of Coring Sample Recovery

CORE

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How Is a Core Cut?

Core Head

- Typically PDC
- No centre

‘Lighter’ Parameters

- Low WOB
- Reduced flow rate

Core Assembly

- Twin barrels – Inner/Outer
- Catcher



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The Coring Goals

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Coring Goals

Primary

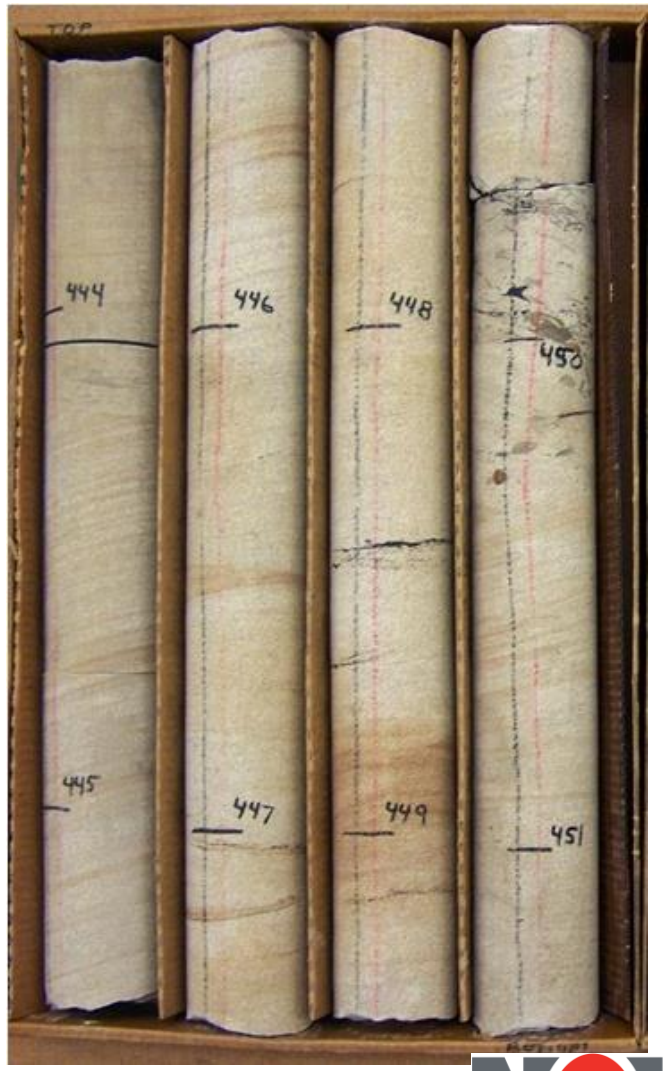
1. Maximise core recovery

- Identify reservoir top
- Achieve 100% recovery
- Maximise coring efficiency

2. Minimise damage

- Mechanical
- Invasive
- Thermal
- Chemical

3. Do so Safely



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Coring Goals / 3 Key

Maximum Recovery

+

Minimum Alteration



Achieved Safely

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The Objectives of Coring

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Wellbore Technologies

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Why Core?

Core analysis generates data used to:

- Calibrate logs
- Aid and confirm log interpretation

- Characterise reservoir
- Provide R_w, R_t, S_w values
- Calculate reserves

- Support development plan
- Select tertiary production fluids

Logs

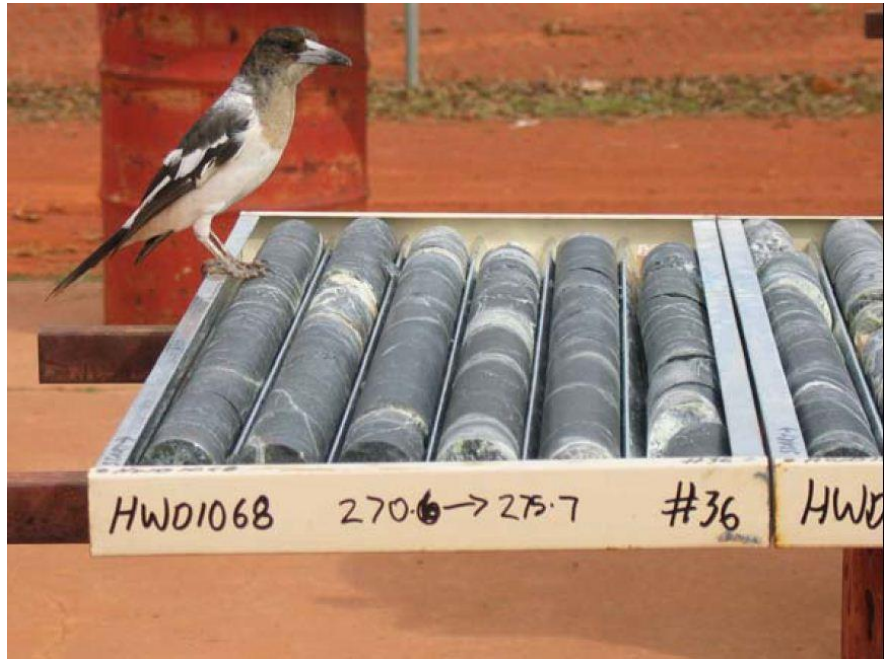
Reserves

Production

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Routine Core Analysis

- Gamma Ray
- Porosity (Storage Capacity)
- Air Permeability (Flow Potential)
- Grain Density (Size, Distribution)
- Fluid Saturations
- Lithology (Formation Type : Limestone/Sandstone, etc)



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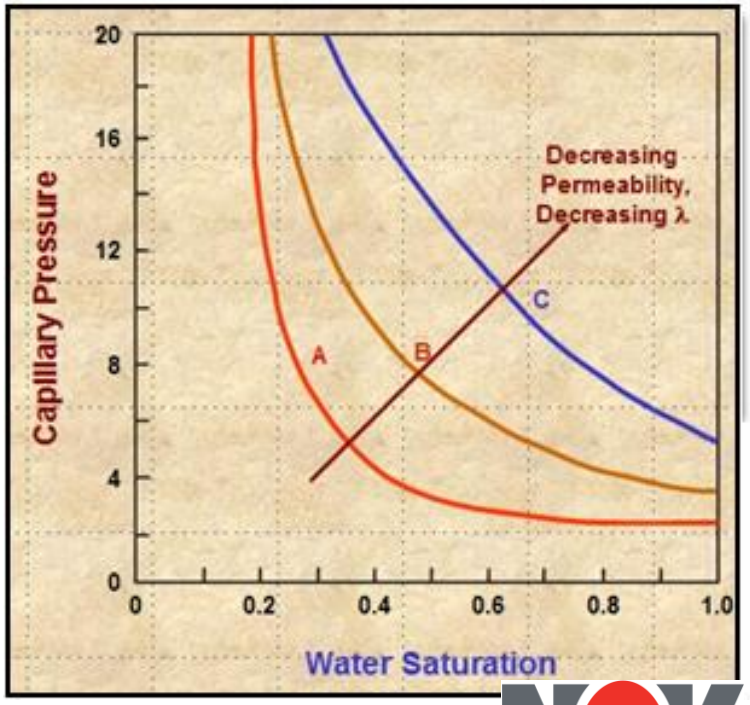
Special Core Analysis

- Wettability
- Capillary pressure
- Relative permeability

- Formation Resistivity Factor
- Resistivity Index

- Sedimentology
- Petrology
- Fracture analysis





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The Coring Type Options

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Core Application Types

Conventional Coring	Standard applications
EOS coring	Fluid analysis
Pressure coring	Fluid/Gas Properties
Wireline coring	Multiple cores
Oriented coring	Directional studies
Sidewall coring	Contingency

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Conventional Coring

Conventional Coring

Standard applications

Top-sub – LAS

Swivel

Stabilisers

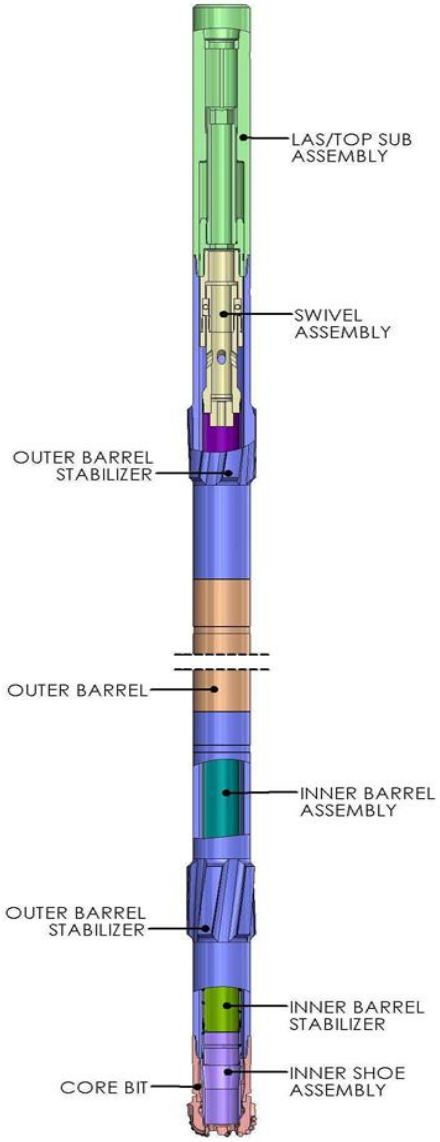
Outer Barrel

Inner Barrel

Stabilisers

Core catcher

Core head



- Connection to string – Adjustment of lead

- **Isolate inner barrel**

Transmit WOB & Torque

- Minimise sticking risk
- Centralise core

Retain & protect core

- Centralise core
- Minimise jamming risk

Catch and retain core in barrel

Cut core

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CONVENTIONAL CORING TOOLS “DURACLOSE” VIDEO

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Enhanced Oil Saturation (EOS)

EOS coring

Fluid analysis



The patent-pending EOS (enhanced oil saturation) coring system provides a means of recovering the oil liberated from the core while tripping to the surface.

A highly absorbent *Wilson* ™ liner made of the latest technology polypropylene non-woven felt is contained within the inner barrel. After the core enters the barrel, the core is in capillary contact with the absorbing material, where oil leaving the core is captured and retained in the *Wilson* liner for subsequent laboratory analysis. A direct measurement of the oil volume in the liner, when added to the core oil volume, provides a correlation to the total oil remaining in the reservoir.

High Capacity – Highly permeable, porous material recovers high oil content

Precise Attribution – Oil is retained in the liner material immediately adjacent to the core section it evolved from, allowing precise location attribution

Resilient Material – Polypropylene Wilson liner does not tear, shred, or delaminate like sponge medium

Easy Identification – Liner does not fluoresce under UV light, allowing easy identification of oil

Accessible – Polyurethane shell is easily opened for evaluation and sample extraction

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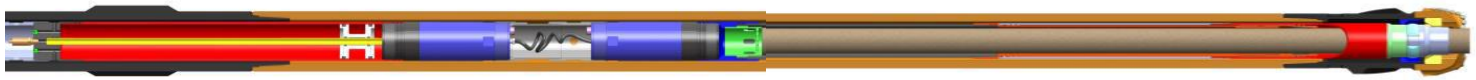
Pressure Coring

Pressure coring

Fluid/Gas Properties

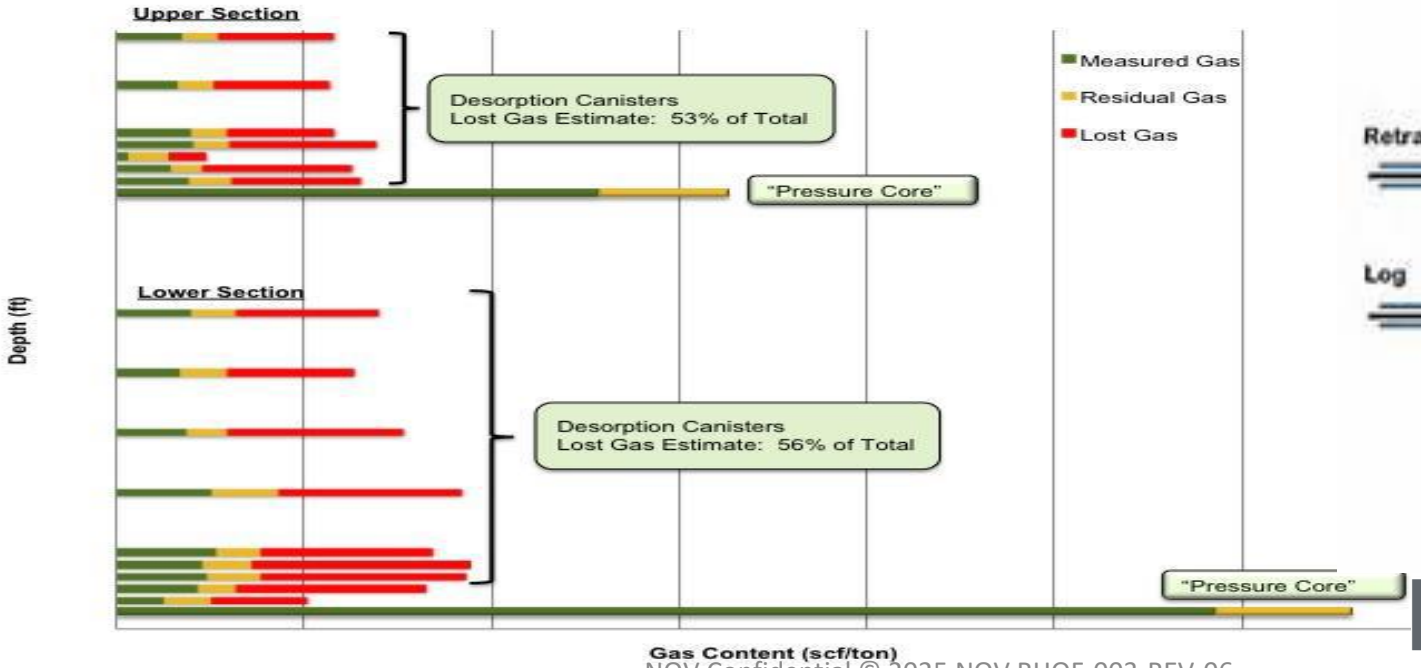
Pressure Storage Canister

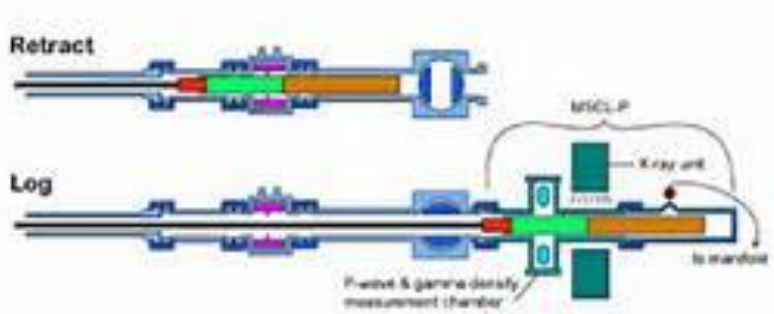
Inner Core Barrel – Core HEad



P / T Recording Modules

The tool has been designed to obtain a direct measurement of total gases and fluids by encapsulating the core in a sealed pressure barrel with a series of canisters that are attached to the barrel. The canisters capture all expanding gases and fluids at safe working pressures.





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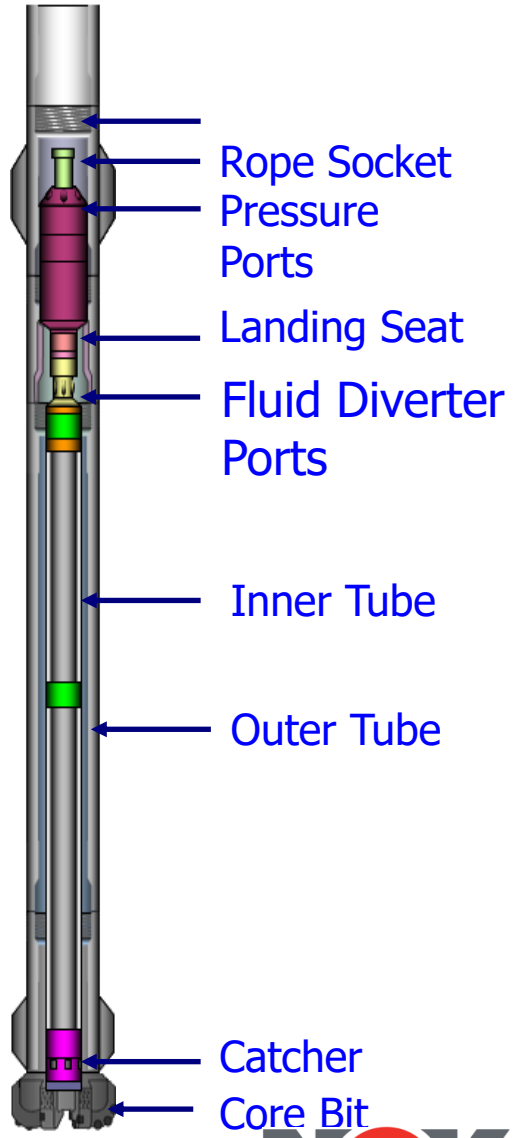
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Wireline Coring

Wireline coring

Multiple cores

- Cores retrieved by wireline
- Barrels pumped down
- Many trips eliminated
- Jamming is only… ‘inconvenient’
- Drilling insert allows drilling



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Corion Express Premium Wireline Coring

www.nov.com

CORING SERVICES



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NATIONAL OILWELL VARCO

Oriented Coring

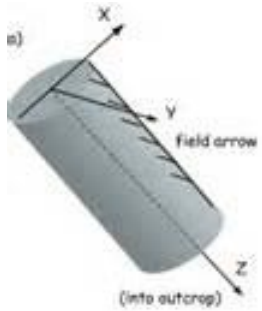
Oriented coring

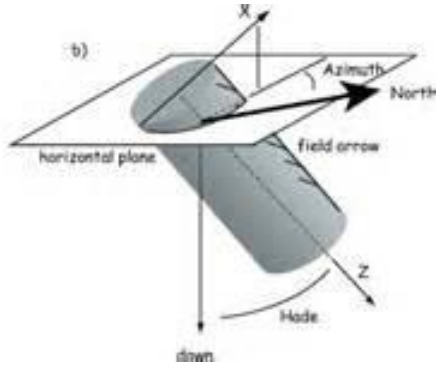
Directional studies

A Core that can be positioned on the surface in the same way that it was arranged In the bore hole before extraction.











Oriented' core has an 'orientation mark' ('OM') along the core marking either the lowermost or topmost line along an inclined drillhole (called the 'bottom mark ('BM') or 'top mark', respectively).

To indicate geographic or magnetic North and amount of "dip" / direction of "tilt", thus facilitate measurement of directional properties in the rock. Orientation used to measure large scale features such as bedding, fractures, flow textures, etc. Recently been used to establish the directions of downhole stress and strain fields.



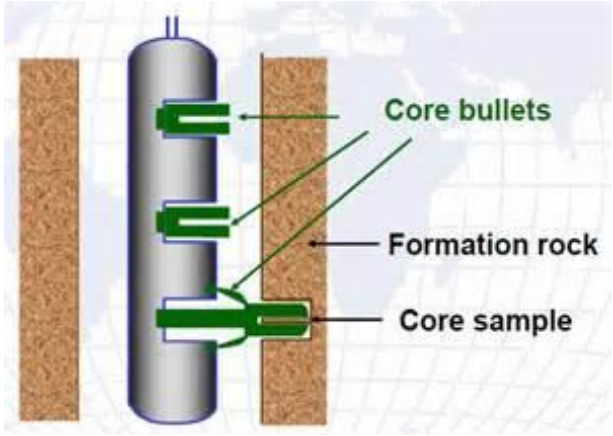
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Sidewall Coring

Type Of Sidewall Coring :

- Percussion Side Wall Coring
- Rotary Sidewall Coring
- Sidetrack Coring System



The tools can be precisely positioned in zones of interest using gamma logs or SP logs as guides. Percussion or rotary drilled samples provide small bits of formation material, suitable for geological and engineering studies. Other systems such as the new Sidetrak Coring System® cut a more conventional type of core angled off to the side of the hole





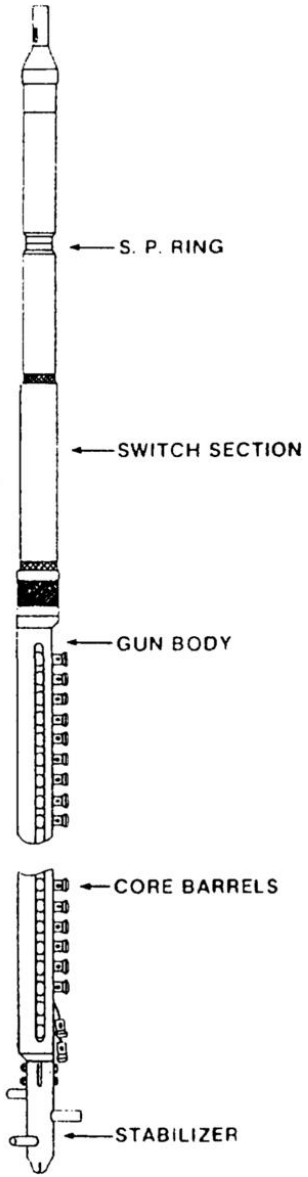
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Sidewall Coring

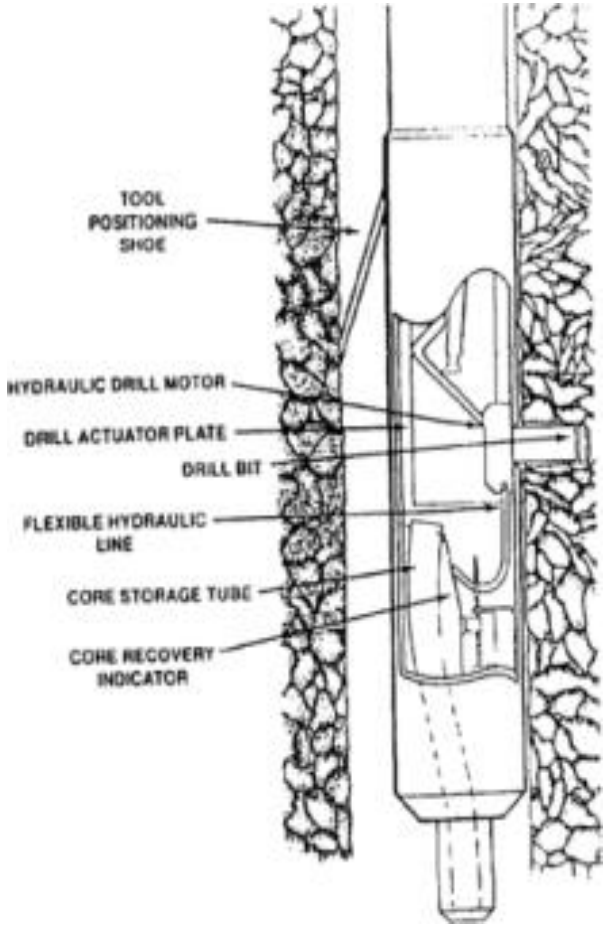
- https:/Side Wall Coring Video

30 Samples
1” OD
1.75” of Length

Suitable for Soft



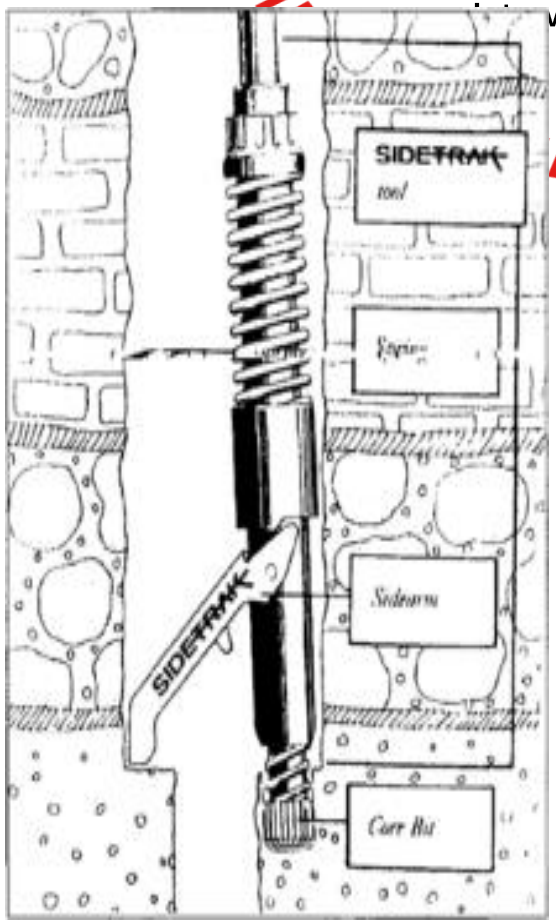
Percussion Sidewall Coring



Rotary Sidewall Coring

30 Samples
0.94” OD
1.75” of Length
Suitable for Hard to friable Rock

Drilling and
vention



Sidetrack Coring

2.5” OD
10ft of Length

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Coring Complications Oil and Gas

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Coring Complications

Tripping Issues
- Potential Swab and Surge
- Potential Fill
- Gas expansion Timing Required

Coring Issues
- Core Breaking in the bit
- Core Jamming induced Milling
- Vibration induced Jamming

Core Quality
- Invasion
- Chemical damage
- Mechanical damage
- Thermal damage

Troubleshoot - Tripping Issues
- Control Trip Out BHA Timing, Circulate, Check Mud Efficiency
- Coring as per procedure, along with Break out core procedure.
- Coring System by default expand gas, Control Trip Out BHA Timing

Trouble Shoot - Coring Issues
- Coring with Optimum Parameter and Break core as per procedure, by pulling BHA very slowly.
- Core Head Selection and Coring Parameter to have optimum penetration rate to minimize core jamming.
- Use of Stabilier, Moderate dogleg severity, Optimum Coring Parameter

Trouble Shoot - Core Quality
- Low Flow Rate, Usage of Low Invasion Design Core Head, Mud properties
- Mud Properties, Correct Preservation Type
- Coring Type Selection, Optimum Coring Parameter, Good Surface Handling, Preservation
- Depends on Well Profile, Mobilise Core to Lab soonest.

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Quick Reference :

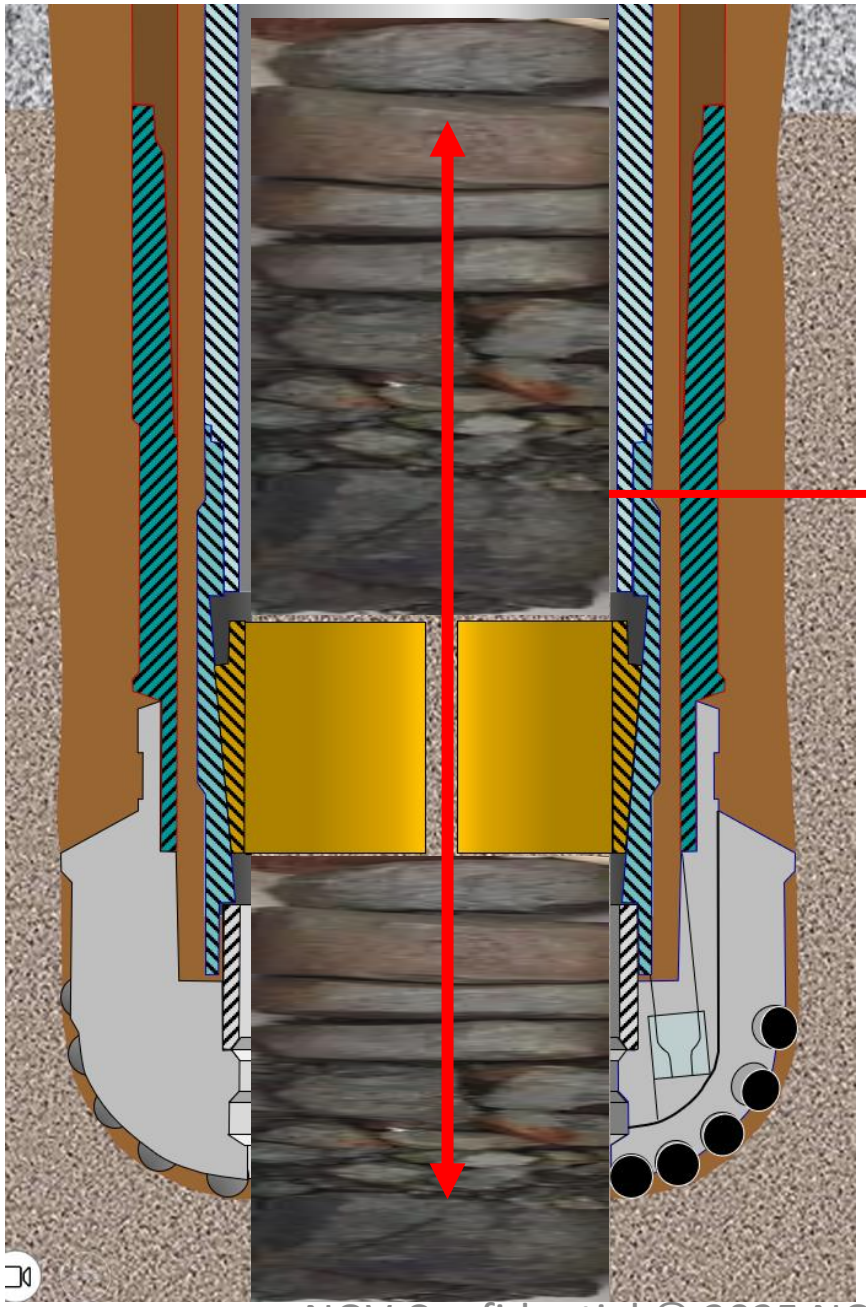
Sudden Pressure Increase	Gradual Pressure Increase	Gradual Pressure Increase	Pressure Decrease	Gradual Pressure Decrease
O ring groove cut into nose of core head due to wear. Blockage of face discharge ports in core head	Core head damage. Pressure caused by O ring groove starting on core head face, becoming deeper, cutting off the fluid courses. Torque will also increase as the waterways plug off.	Indication of jammed inner barrel. Pressure fluctuation due to the jammed core taking weight, causing the core head to drill off and stop or reduce penetration rate.	Indication of jammed core barrel. The core cannot enter the inner barrel, causing the bit weight to be transferred to the core, which then stops or reduces penetration.	Indicated by standpipe pressure being lower than the off-bottom pressure. Possible hole in drill string, or washout in thread connection of drill string or core barrel.

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Coring Complications Geothermal

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Rocks at Geothermal are different than at Oil and Gas.

It mostly in boulders that could be brittle and hard at the same time.

Thus Core Jamming is an issue.
And also core falling down while POOH, due to non cemented profile of formation

Note :

Core Jamming and Milling at Geothermal formation is undetected at Surface Gage, due to highly permeable, penetration could still be present.

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Coring Geothermal Issues

	Geothermal Characteristics Hazards	Issues Induced While Coring
1	High Temperature	- Lack in Inner Barrel spacing calculation - Core Jamming
2	High Torque	- High Vibration - Core Jamming
3	Brittle Hard Permeable Formations	- Falling of Core sample while POOH - Wash Out Core - Core Jamming
4	Highly Fracture and Highly Permeable formation	- Core Jamming - Wash Out Core



All above relates to Low Core recovery to Zero core recovery

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Geothermal Issues Consequences - Mitigation

	Hazard / Issues	Mitigation
1	High Temperature – Lack of Spacing Calculation	- Use of Steel Inner Barrel if Temperature >205degC – 350degC - Use of Aluminum if temperature only until 205degC
2	Core Jamming	- Use of Anti Jamming System (NOV JAMTEQ) - Use of Low to Moderate Coring parameter
3	Washed Out Core	- Use of Low parameter - Use of Less Aggressive Core head (i.e 7 bladed or 9 bladed core head) combined with Shaped cutter like 3D ION Cutter and 4D ION Cutter
4	Core Falling off while POOH	- Use of low Flow rate while POOH - Longer Trip Out Time - Use of Combination Duraclose and Jamteq

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The Coring Process

DOWNHOLE

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Drilling and Coring Agenda

DRILLING



- On-Bottom & Less BHA Trip
- Reach TD
- Maximum ROP

CORING



- All Target & More BHA Trip
- 100% Recovery
- Optimum ROP - Minimal Damage

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The Coring Process

- Safety (Planning and Meeting)
- Identify the target zone
- Prepare and assemble equipment

Preparation

- Run in the hole
- Initiate coring process
- Core
- 'Catch' core
- Retrieve safely

Coring

- Process on Surface
  - Measurement/description
  - Cutting
  - Treatment/Preservation

Processing

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Coring Flow System in General

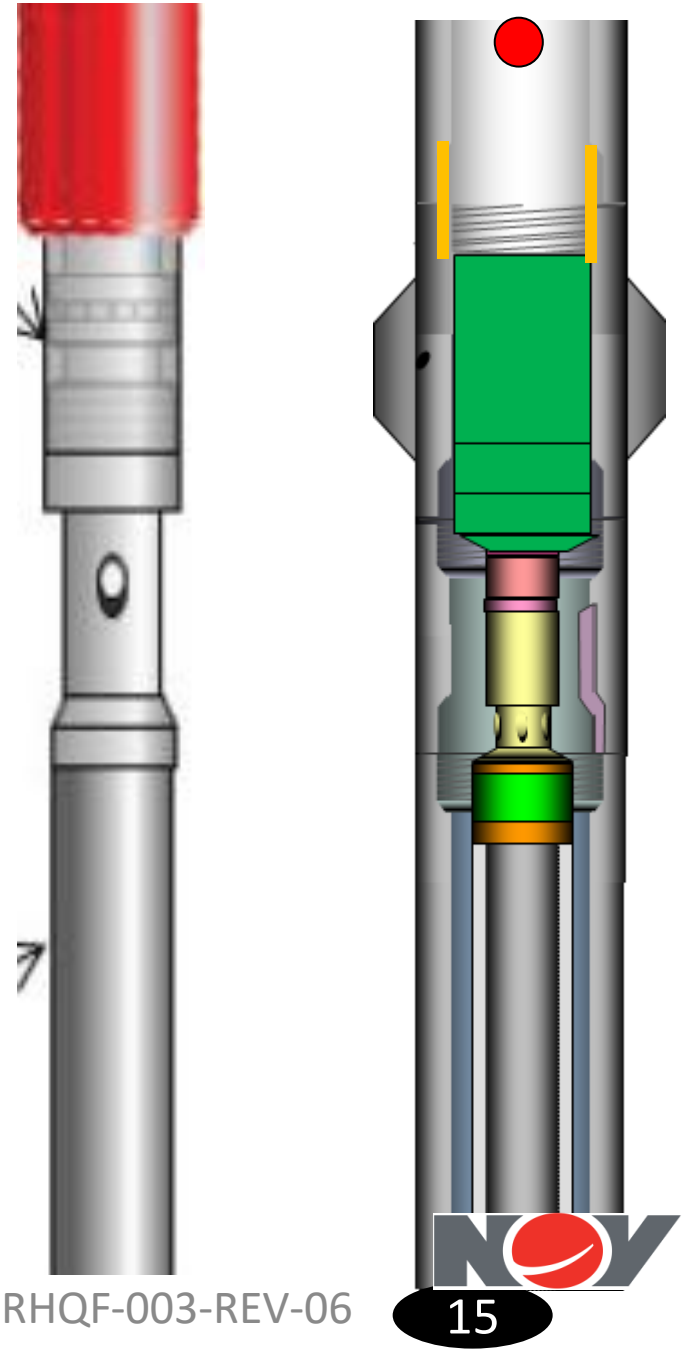
NOV Wellbore Technologies
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NOV Energy Products and Services

Swivel

- Prevent inner core barrel rotating with the outer core barrel.

Procedure

- Circulate through Inner Barrel when tripping in for bottom
- Clean Bottom Hole before beginning of coring.
- Drop Ball and start coring
- Venting of gas when tripping out



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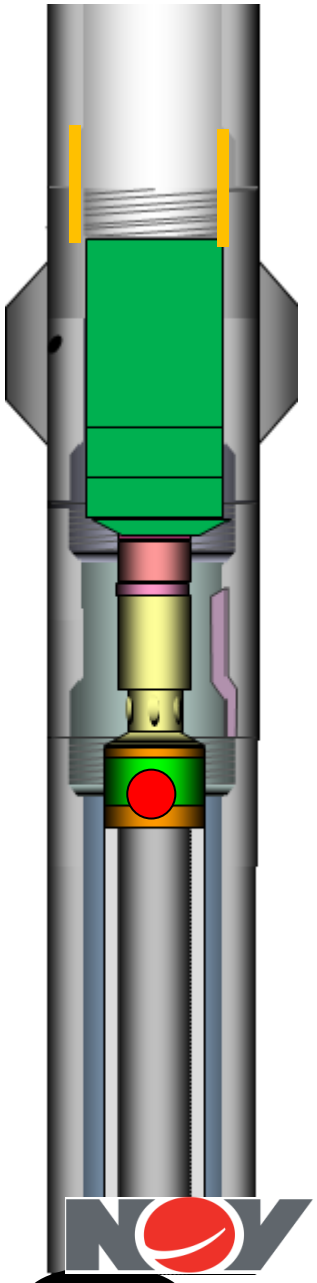
Swivel

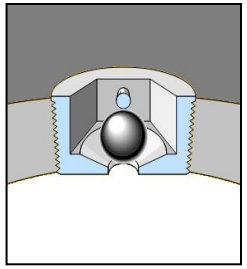
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Procedure

- Circulate through Inner Barrel when tripping in for bottom
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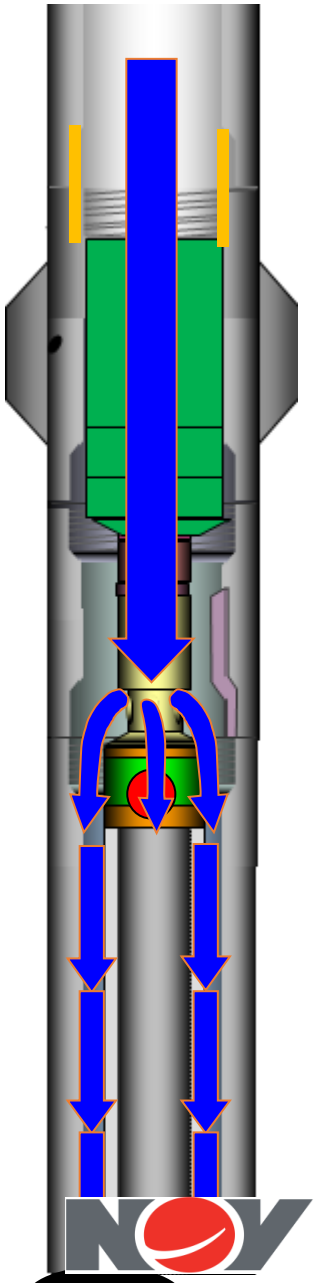
Swivel

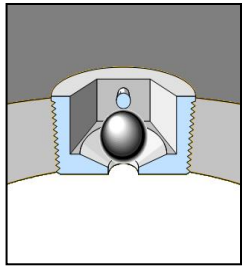
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Procedure

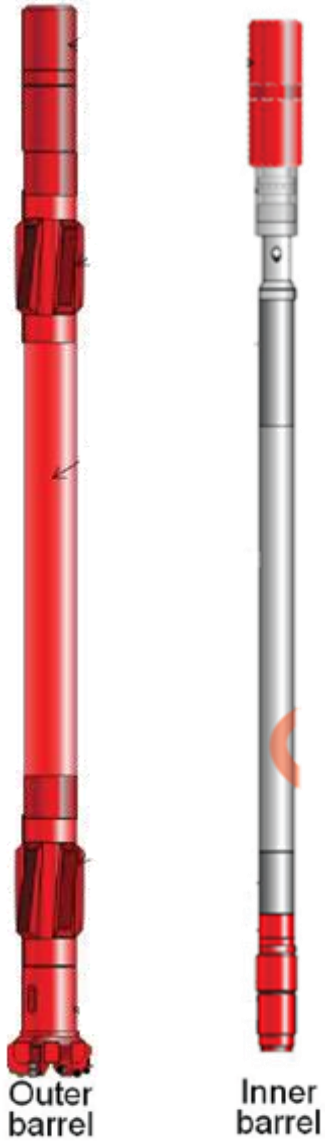
- Circulate through Inner Barrel when tripping in for bottom
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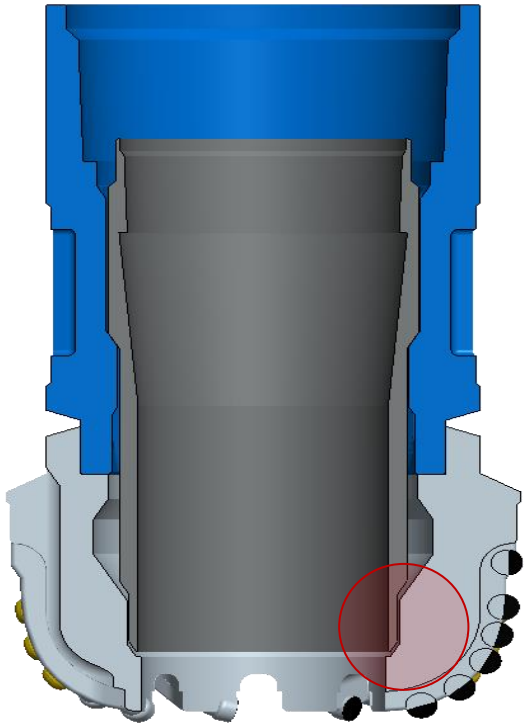


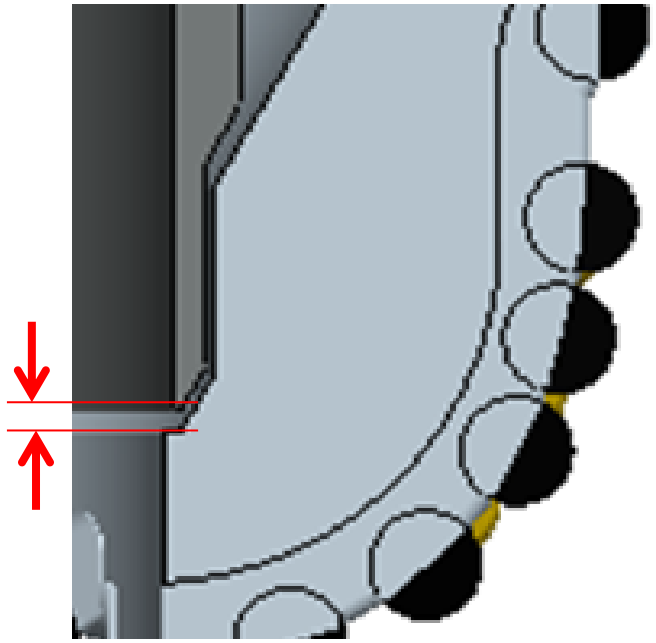


Main Components



Distance between the core bit throat & lower part of the inner core barrel shoe.





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How Core Catcher Spring Works

Animation showing the core coming inside the conventional core catcher-inner barrel (while coring)





Drawings and animation done by Carlos Pereira

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CORE HEAD





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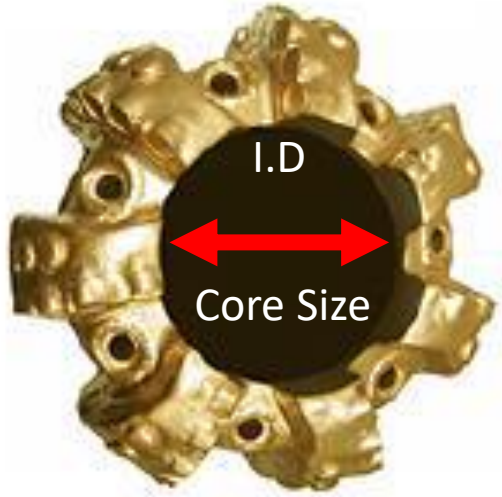
Two Key Dimensions

- **Outside diameter** – Hole size
- **Inside diameter** – Core size

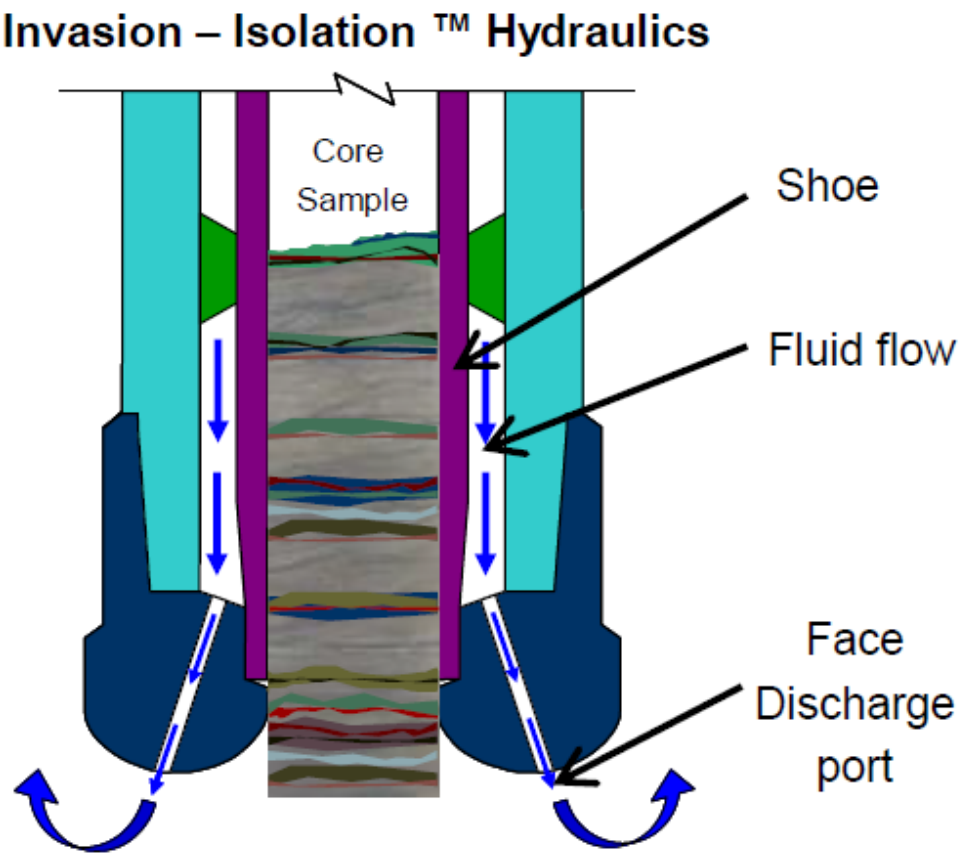
Typical Dimensions

Bit Size	Core Size
12 ¼"	5 ¼"
8 ½"	4"
6"	2 5/8"





Low Invasion Design



Invasion – Isolation™ Hydraulics:

- Flow ports direct fluid flow away from the core
- Customized bit designs seal off mud flow to the core by incorporating;
  - Lip ring in combination with shoe
  - Long nose shoe
- Flush inner diameter further restricting mud to pass through the bit.
- Invasion – Isolation™ Hydraulics provides
  - More accurate fluid saturation analysis
  - Minimize filtrate invasion
  - Enhanced core recovery

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Core Head Performance - Abilities

- Drillability
- Durability
- Stability 
- Steerability
- Adaptability
- Compatibility 
- Availability
- Reliability
- Acceptability
- Repairability 
- Retrieveability 

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Core Head Types – Relative Benefits

- **Natural Diamond**
  - Durable – hard rock
  - Not aggressive

Disadvantages

  - Low ROP
  - Prone to balling
  - Core invasion

- **PDC**
  - More versatile
  - Higher ROP
    - Better core
    - Lower rig costs
  - Avoid balling
  - Low invasion

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Lithology - Guidelines

For Core Head Design in General



Soft

- Claystone
- Shale



Medium

- Chalk
- Siltstone



Hard

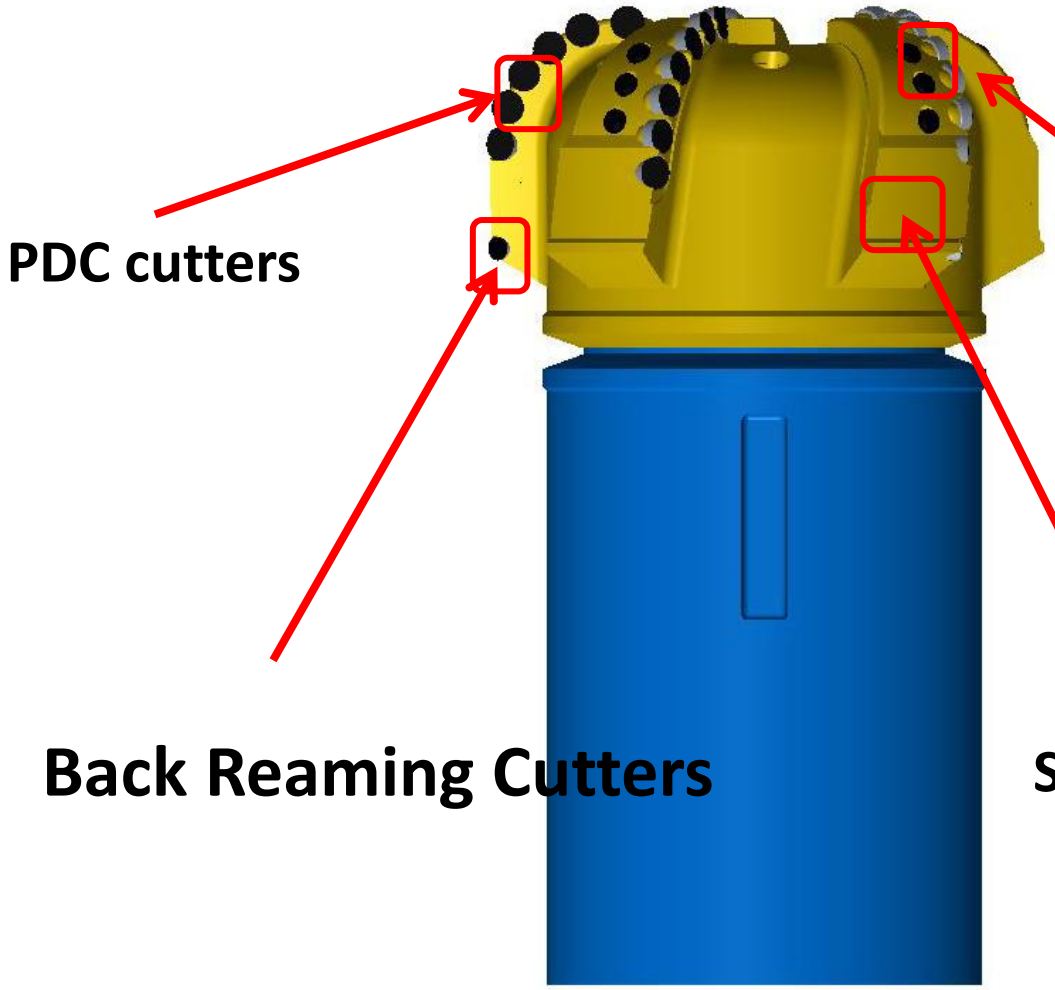
- Sandstone
- Crystalline limestone

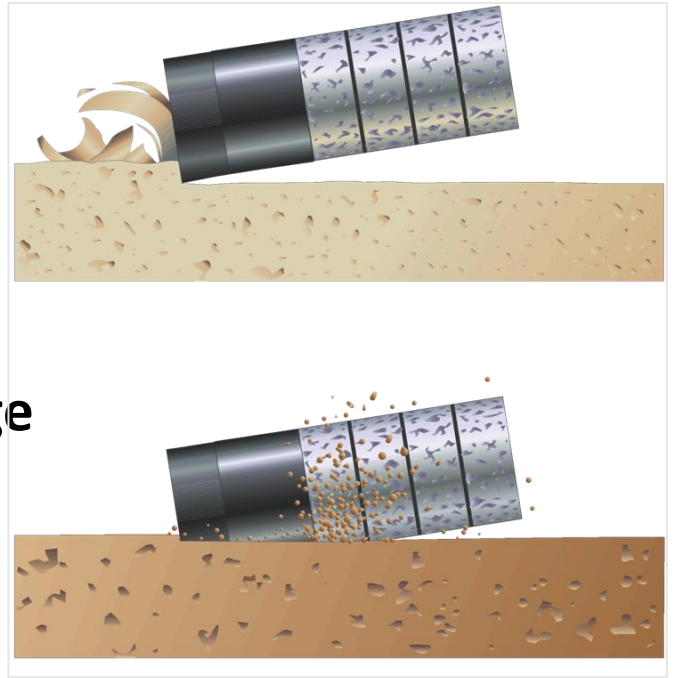
- Few blades
- Few cutters
- Large cutters
- Small back rake
- Long tip profile

- Many blades
- Many cutters
- Small cutters
- Large back rake
- Short tip profile

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PDC Core Head Features





- Impreg Diamond Core Head



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OTHER TYPE OF CORE HEAD

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CB Series – Natural diamond

Natural diamonds



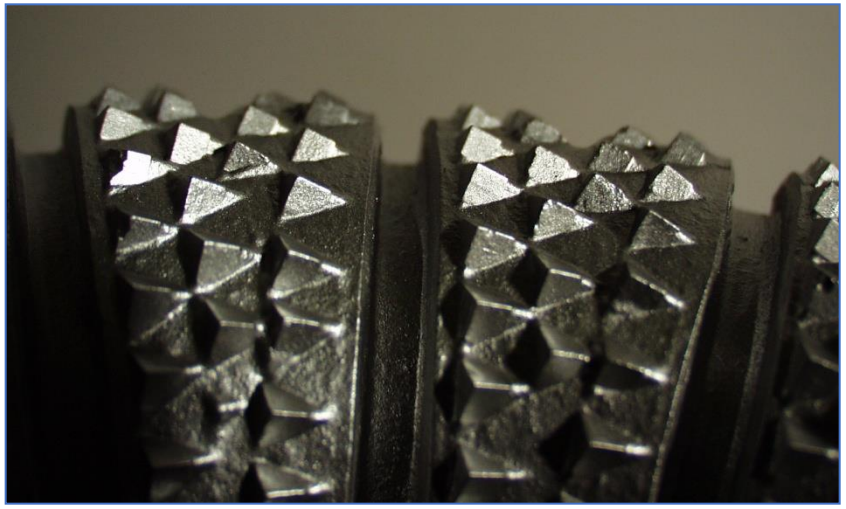




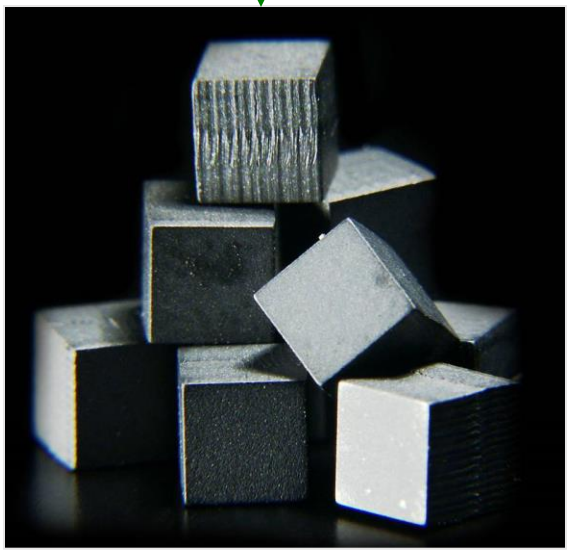
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CT Series – TSP (Thermally Stable Polycrystalline)



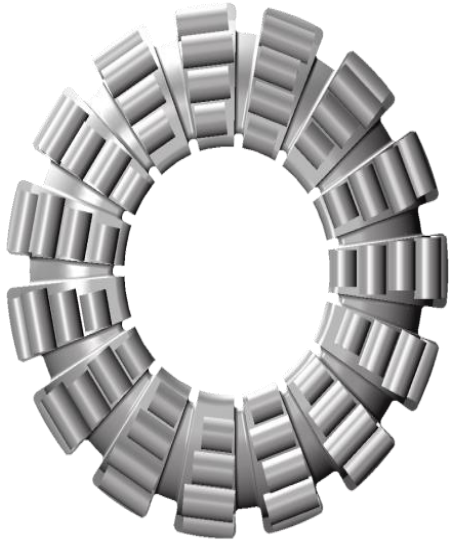


TSP



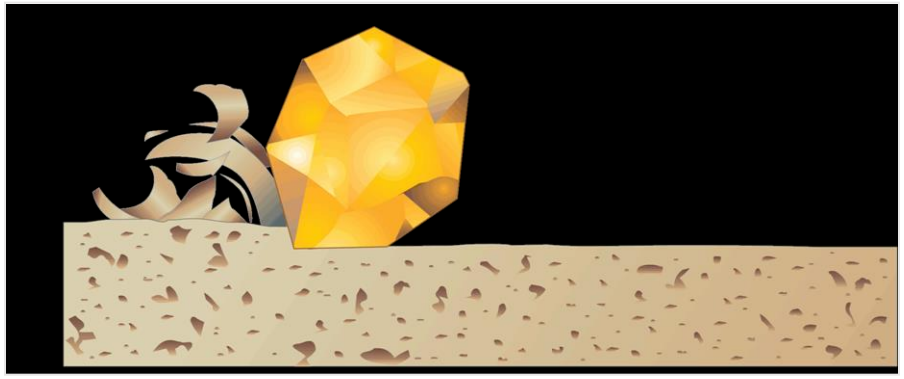
Full Diamond Impregnated

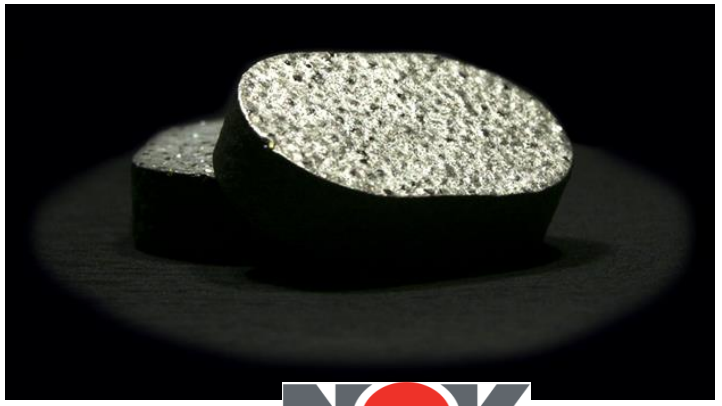






The block concept





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PDC Core Head with 3D ION cutters PHOENIX Core Head



Product Report



8 ½" x 4" CDPFX913S-A1

Design Features of this Core Head

DuraForceXD™ Impact Resistant Cutters Designed for maximized impact toughness. The cutter incorporates new polycrystalline diamond material chemistry specifically selected for impact toughness and a patented deep-leached thermostable surface layer for unrivaled abrasion resistance, allowing the bit to drill considerably further and faster to improve overall bit performance.

Design Specifications

System Type:	
Core Dia:(ins)	4"
Blade Count:	9
Make up Length (ft):	1.24
Connection Type:	Durateq
Shank Diam (ins):	7.000
Cutting Profile:	
Hydraulic Style:	
Low Invasion:	
Shank TFA:	
JSA (in^2):	8.74
Body Material:	Steel
Gauge Geometry:	Spiral-Trailing
Plug Bit	
Part ID:	

Contact your ReedHycalog Representative for recommended operating parameters in your application. ReedHycalog reserves the right to revise these specifications, based on advances and improvements in technology.

This report is valid for 30 days from 03-Dec-2013







Product Report



8 ½" x 4" CDPFX713S-A1

Design Features of this Core Head

DuraForceXD™ Impact Resistant Cutters Designed for maximized impact toughness. The cutter incorporates new polycrystalline diamond material chemistry specifically selected for impact toughness and a patented deep-leached thermostable surface layer for unrivaled abrasion resistance, allowing the bit to drill considerably further and faster to improve overall bit performance.

Design Specifications

System Type:	
Core Dia:(ins)	4"
Blade Count:	7
Make up Length (ft):	1.24
Connection Type:	Durateq
Shank Diam (ins):	6.900
Cutting Profile:	
Hydraulic Style:	
Low Invasion:	
Shank TFA:	
JSA (in^2):	10.24
Body Material:	Steel
Gauge Geometry:	Straight
Plug Bit	
Part ID:	

Contact your ReedHycalog Representative for recommended operating parameters in your application. ReedHycalog reserves the right to revise these specifications, based on advances and improvements in technology.

This report is valid for 30 days from 04-Feb-2014

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PDC Core Head - 4D ION

-→ (699)4D ION VIDEO



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ION™ 4D Shaped Cutter Technology

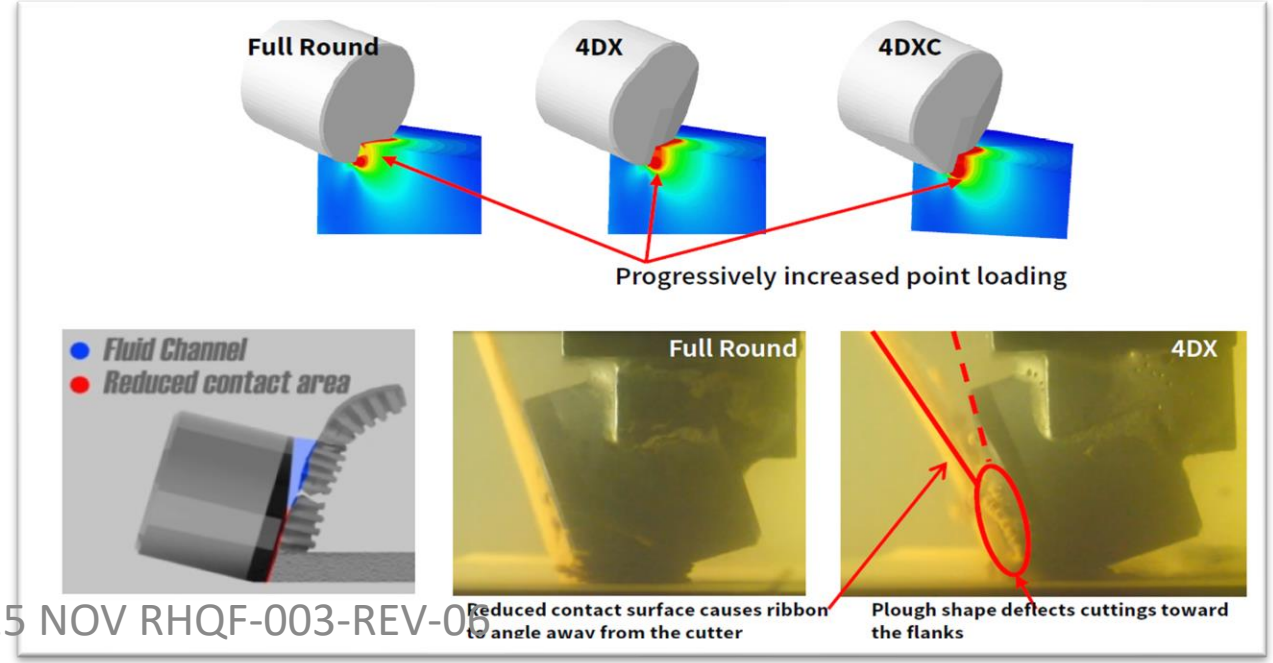
Our patented, flagship 4D shaped cutter is the fastest and most efficient geometry developed by ReedHycalog and is backed by proven field record-breaking performances.



4D 4DC 4DX 4DXC

ION™ 4D & 4DX application-specific shaped cutters have been developed to further improve drilling efficiencies, both reducing torque while increasing rate of penetration (ROP). The 4D nonplanar cutting face creates a plough effect which increases point loading in the shearing direction while reducing drag, and creating smaller cuttings.

ION 4DC & 4DXC shapes couples 4D technology with proven 3D (chisel) technology. ION 3D chisel cutting-edge profile produces point loading in the axial (WOB) direction improving fracture propagation and further lowering mechanical specific energy through torque reduction while achieving much higher ROP.



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Core Head – No Go RING Gaged before Shipment







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Core Head – No Go ID Throat Gaged before Shipment :





Core Head – Double Checked Nozzle Gaged before Shipment :





COFFEE BREAK – 15 Minutes

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INNER BARRELS

- **3 Types of Inner Barrel**
    - 1. Steel Inner Tube
    - 2. Aluminum Inner Tube
    - 3. Fibreglass Inner Tube

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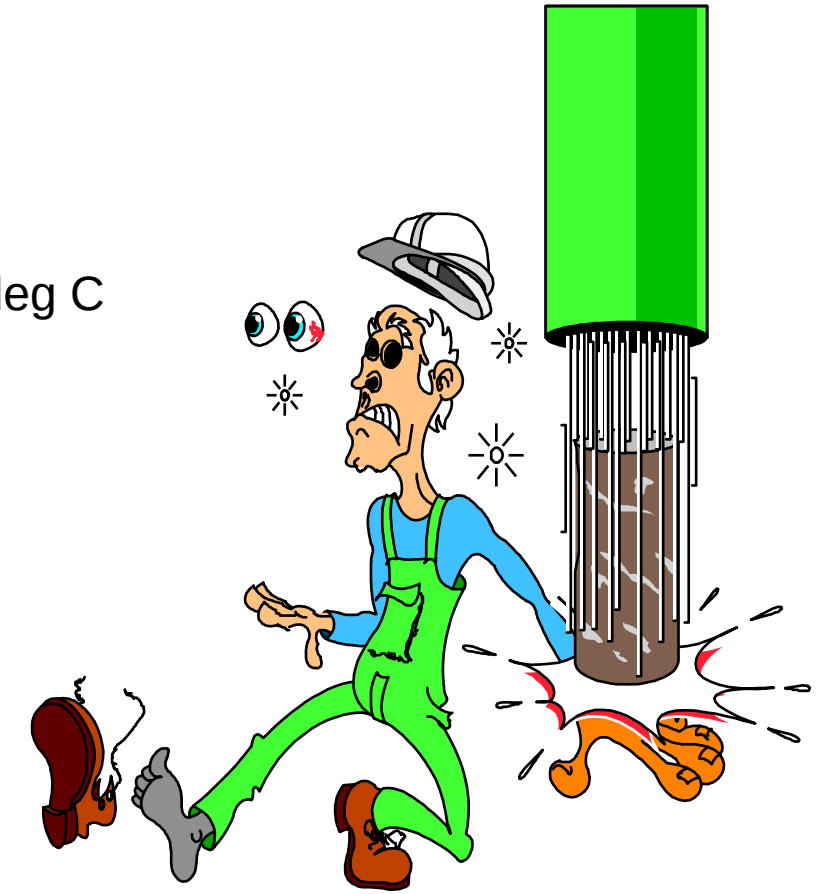
Inner Barrel - Steel



- Mainly available in market for a 4” Core size
- Used for very High Temperature Exceed > 205 deg C – 300 deg C

- Usually single Tube System, Using Core Tong Boot and Handle to Extract the core at Rig Floor
- Complicated handling at surface

- **Nowadays, a “Triple Tube System” were made, An “Aluminum Liner” is inside a Steel Inner Tube, on which the Steel Inner Tube is inside a Core Barrel**



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Inner Barrel – Fiberglass Tube

Inner Barrel – Fibre Tube ™

Fiberglass Inner Tubes are designed to significantly improve core quality and recovery. With Fiberglass Inner Tubes coring is safer, more efficient, and the core is more easily handled. Rig site core gamma logging is also possible.

The disposable Fiberglass Inner Tubes become the packaging system for the core from the time it is cut until it reaches the core analysis laboratory. These inner tubes offer little resistance to core entry which reduces core jamming and loss, and significantly improves core recovery and coring efficiency.

COLLAPSE

	MPa	psi
2”5/8	16.3	2364
4”	10.5	1522
5”1/4	4.4	638







Dimensions :
30ft Length
Available in 3 sizes Sizes: 2 5/8”; 4”; 5 ¼”
<120deg C (Recommended Max Operating temperature)
Collapse psi <1,800psi

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Inner Barrel - Aluminum

<u>Inner Barrel – Aluminum SuperGlide™</u>

- Available for 3 sizes

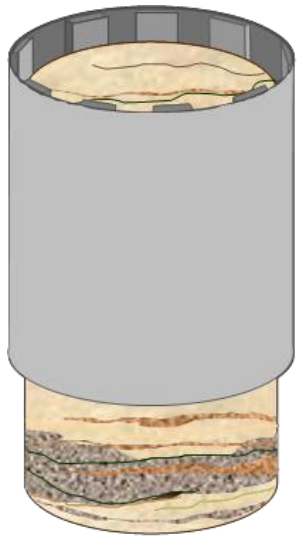
  Sizes: 2 5/8”; 4”; 5 ¼”

- Collapse pressure of 2,983psi

- 30ft lengths – plus 1ft extensions – 30ft (nominal 9m)

- Gas passages, aides mud bypass, centralizes & supports core effectively.

- Rec. Max Operating temperature <205deg C





Tube diam.	Type	Burst pressure (MPa)	Collapse pressure (Mpa)
2 5/8"	Slick	27.2	61.5
2 5/8"	Fluted	21.8	48
4"	Slick	14.9	25.9
4"	Fluted	19.1	22.4
5 1/4"	Slick	16.3	32.3
5 1/4"	Fluted	16.1	26.8

Note : Mpa x 145 = psi

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Background - Length Adjustable System (LAS)

Safety Joint Asembly

LAS allows lead adjustment without breaking inner or outer barrel connection.



Shims

Now



Traditional the length was adjusted by using shims, placed around the swivel area.

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Length Adjustment System (LAS)



1 turn (360°) of the LAS mandrel
- Easy Spacing at Surface

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SPACING ADJUSTMENT

TOTAL SPACING = INITIAL SPACING (5mm) + THERMAL SPACING

Spacing is function of:

1. Surface T°
2. Downhole T°
3. Inner barrel length
4. Inner tube material type

Spacing chart available for:

Aluminium

Fiberglass

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Spacing of Aluminum Inner Tubes Based on Thermal Expansion Coefficients

Bottom Hole Temp. Minus Surface Temp.	30 Foot Core Barrel Extra Length Spacing	60 Foot Core Barrel Extra Length Spacing	90 Foot Core Barrel Extra Length Spacing
Degree C	Mm	Mm	Mm
20	2.14	4.28	6.42
40	4.28	8.56	12.84
60	6.42	12.84	19.26
80	8.56	17.12	25.68
100	10.7	21.4	32.1
120	12.84	25.68	38.51

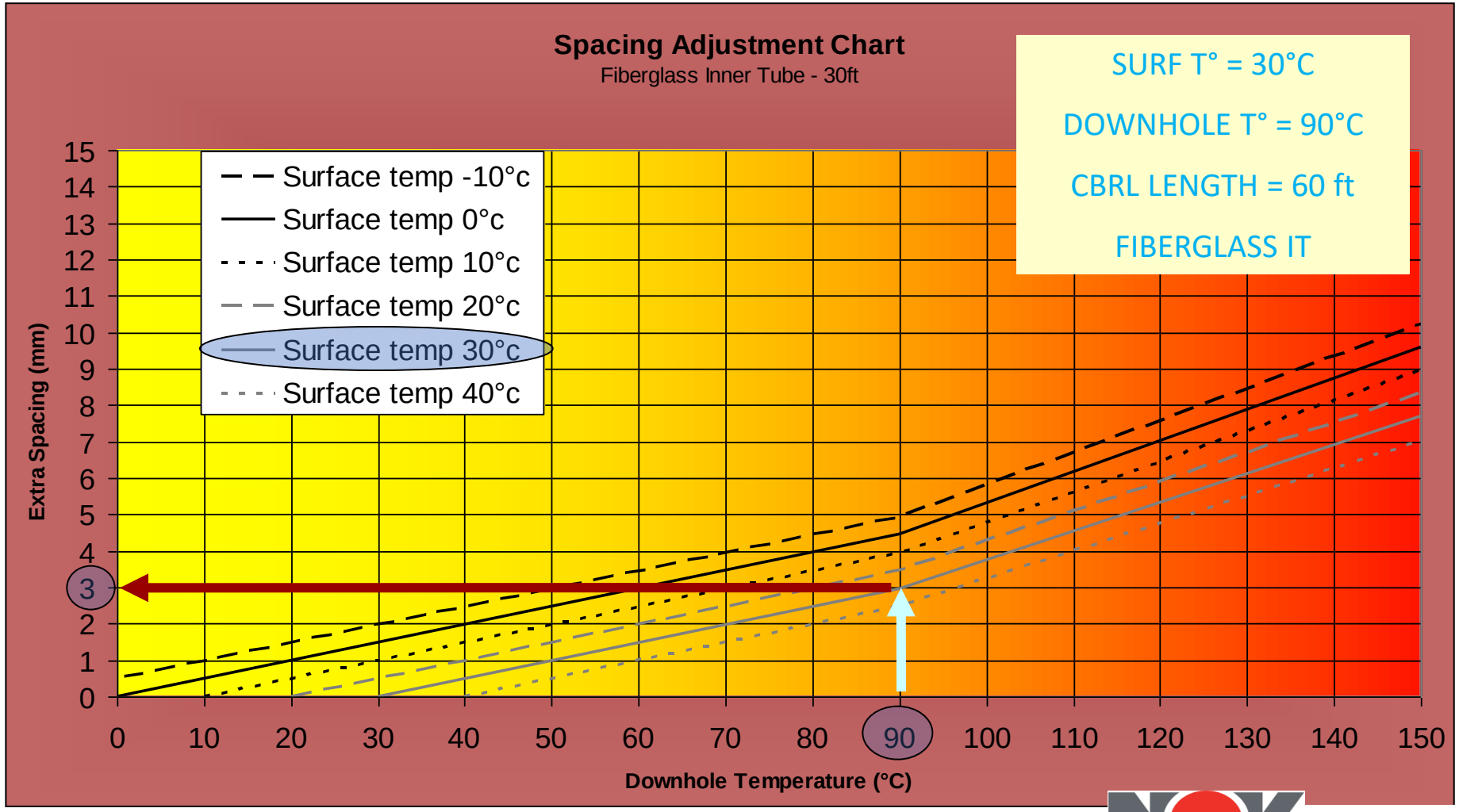
TOTAL SPACING = INITIAL SPACING (5mm) + THERMAL/Extra SPACING

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Spacing of Fibre Inner Tubes Based on Thermal Expansion Coefficients

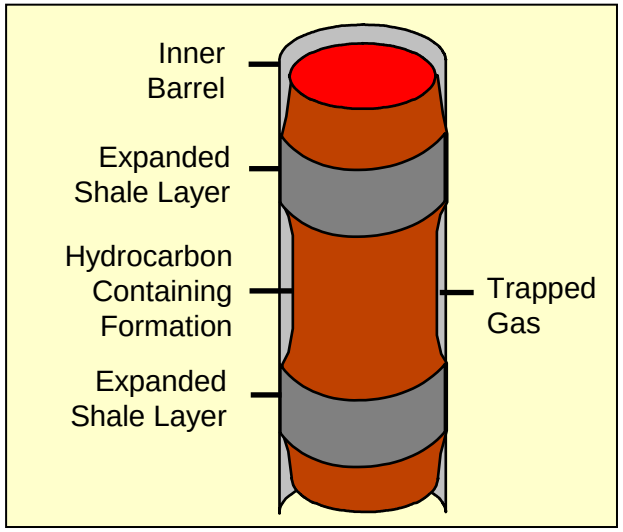
TOTAL SPACING = INITIAL SPACING + THERMAL SPACING

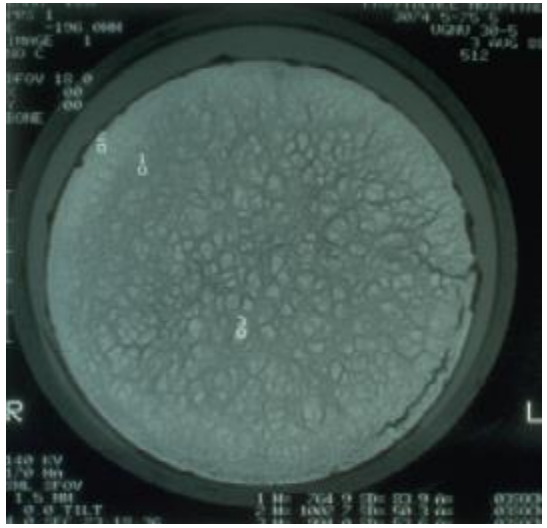
TOTAL SPACING = 5 + (2(If 2 Inner tubes Assy) x 3 (Thermal Spacing Calc. for each IT as per Chart)) = 11 mm



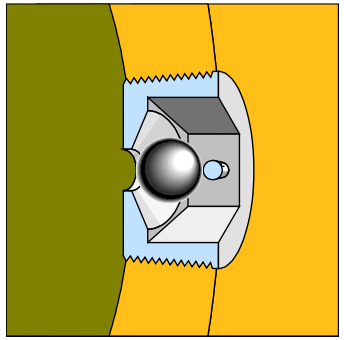
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Vented Valves for Inner Tube (for HTHP Well)





Core damaged by gas expansion



Vented valves:

- 2 ft interval w/ a 120° angle
- 15 per 30ft tube
- appropriate drills - thread taper - allen key - loctite

The valve prevents entry of drilling fluid, but allows exit of gas from IT. During coring, the ball is held in place by the differential mud pressure over the IT.

Shop installation of valves -

NTD 1/8" Pressure Valve

High Temperature Loctite 620 or 648

Drill diam = 8.5mm

Thread TAP 1/8" NPT

Modified Allen Key - T5x200

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STEEL BASKETS\
Forr Inner Barrel Storage and Transport



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Surface Handling



POOH INNER BARREL PROCESS



LIFTING NUBBIN
(Attached to Inner Barrel Box Conn.)

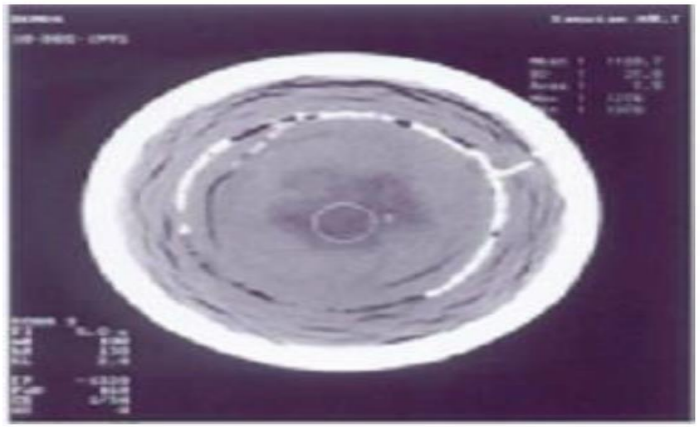


GUILLOTINE BOOT
(USE IF MORE THAN 1 JOINT INNER BARREL)

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Core Breakout – C-Clamp & Pipe Cutter

Back in the days when core is retrieved onto the surface, the inner tube joint are separated by simply applying a breakout torque at the inner tube joints. What this resulted in was torque damage to the core as below:-



Core Damage due to Rotational Breakout



NOV recognizes this problem and have come up with a method to breakout the core without rotating the inner barrels.







Operation: The inner tube is first held together with a C-Clamp and cut with a Pipe Cutter. After that, the shear plate on the guillotine boot is activated by hydraulic pump to provide a smooth cutting process. This approach ensures minimal rotational or lateral stress is imparted to the core during the recovery and laydown process.

Surface Handling



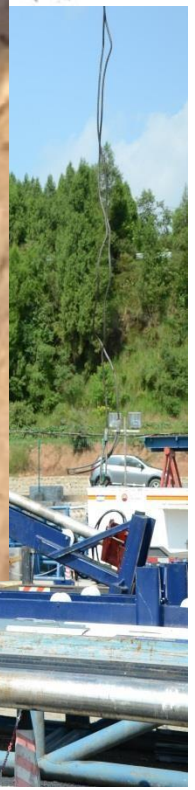


Non Man Riding Laydown Cradle

- Eliminate bending movement of barrel
- Removes Man
- Hydraulically ac
  hold Inner barr









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Core Sample Processing

Measuring and Marking, of Core Sample

- Will be conducted by Geologists Team on board.
- Before cutting process, Core must be handled & prepared properly, together with Coring Engineer



REQUIRED:

- Geologist Team and Experienced Field Technicians
- Order of process – clean, measure and mark, cut & cap.
- Sufficient equipment to continue further process properly

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Core Sample Processing

Air Saw – Cutting Process

- Box style with Diamond & Abrasive Blades







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RESIN LIQUID - Preservation



Uses 2 (Two) Chemicals – Combined Equally
Past Cast A149 (Resin) and Past Cast B149 (Hardener)



Resin both ends, and capped both ends

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RESIN FOAM - Preservation

FOAM INJECTION





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RESIN FOAM - Preservation

Foam Injection Method benefits :

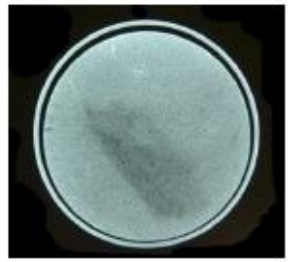
1. **Better and Reliable process**
2. **Non invading to the core matrix**
3. **Invisible to the CT Scan**
4. **Easily removable from the core surface**
5. **Non hazardous chemicals involved**
6. **Minimal core handling**



BEFORE



AFTER INJECTED



CT SCAN

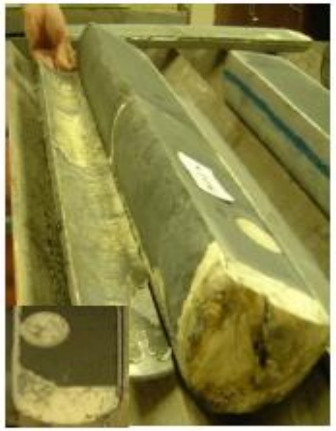




The Foam Injection process can be performed prior to cutting the core into sections **(1m/3ft)**



The Foam Injection process can be performed to a Full Length Inner Barrel **(9m Uncut)**



After Slabbing Process In Lab



Slab and Scan Picture

NOV CORING SERVICES

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HDPE Core Crates

Custom core transport crates are industry best practice for securely moving 1 m containerized core sections to the core analysis laboratory or core store. Our HDPE core crates use a palletized transportation system with supporting profile foam inserts designed for different inner barrel sizes. When combined with a well-considered stabilization method, these crates secure and protect the core samples during transport.

For additional security, shock labels and electronic monitors may be used to track transportation parameters. Shock labels clearly indicate when a potentially detrimental shock has occurred during transport, but do not record a timestamp. Electronic monitors can be set to monitor temperature and humidity events as well as location and standard impact levels, recording the date and time at which each event occurred.



Benefits

- **Core Support –** HDPE core transport crates provide protection and support for the core
- **Safe** – Fold down sides assist in manual handling.
- **Operating Environment –** Can be used in hostile conditions
- **Easy Movement** – Can be easily transported with forklifts once filled
- **Built to Ship** – Standard size for shipping

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Packaging – Wooden Core Boxes

Core Crates

- Profile Cut Foam Padding
- 1m per wooden box



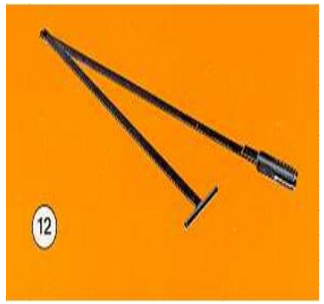
AT WORKSHOP



AT SITE AFTER CORE INSIDE AND BOX LABELED

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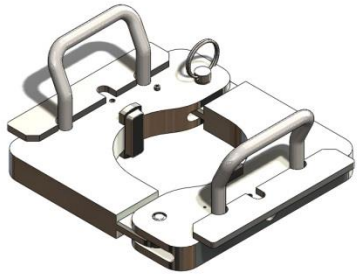
Handling Tools



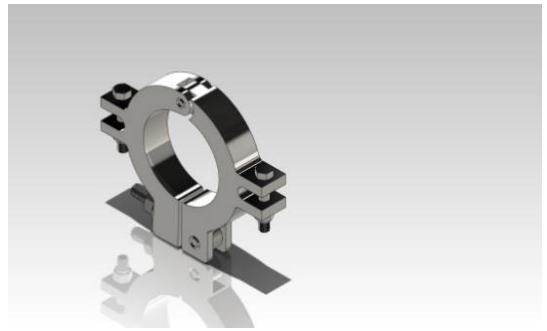
Drop Ball Retriever



Pneumatic Air Saw



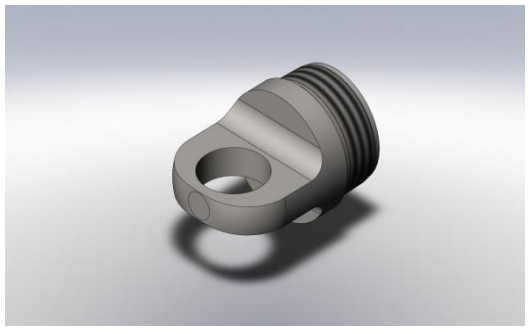
Bit Breaker



Inner Tube Clamp



Guillotine Boot



Lifting Bail / Plug



Electric Saw



LAS Wrench

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ADDITIONAL

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Spot Gamma Logger

Gamma Logger

- Spot Type every 10 to 25cm
- Spectral / Total output (keV)
- Source Cesium137, Calibrated to 662 keV

Measuring the natural Gamma radiation coming from the core in KeV (energy level). Total counts gives you the entire spectrum. U, Th and K gives energy levels from those specific regions of interest.



"...on site core analysis allows the operator to make effective, real-time assessments of the well site."



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Spot Gamma Logger

Gamma ray logs provide valuable insight into the mineral composition of recovered formations. We offer gamma units capable of providing both total and spectral (K, U, Th) gamma logs. Information is presented in both tabular and graphic form, with gamma counts against depth.

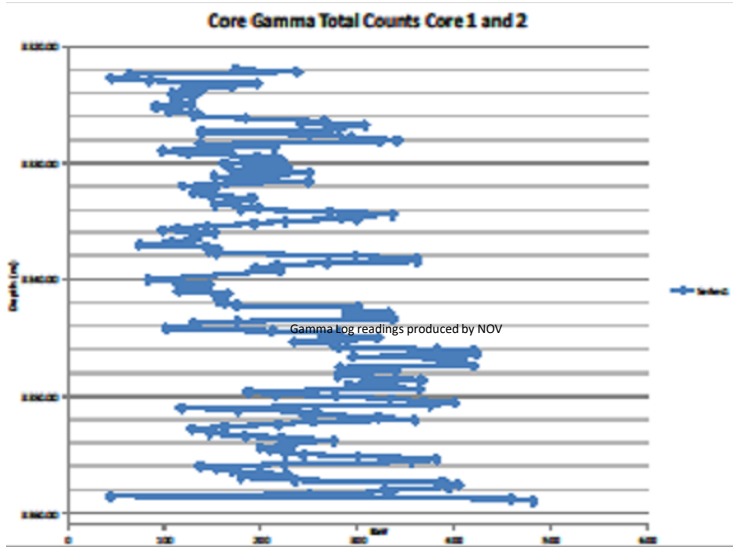
These logs help to differentiate between types of rocks and minerals found, including sandstones, shales, and particularly clay minerals. The generated logs are used to define depth corrections with other wireline logs, as well as to assist with correlations from offset well gamma data to determine specific signatures and thus find expected formation specific markers. These data sets can also assist with critical path decisions, such as whether to core or to drill ahead or when selecting preserved sample or core plugging sites.

Benefits

- **Background Noise Elimination –** Calibrated to ensure location specific accuracy
- **Gamma Counts** – Spectral and total counts are available
- **Sampling Options** – Variable sampling frequency for maximum flexibility
- **Robust Units** – Units are built tough and resilient for maximum portability



Core gamma measurement, NOV coring services provides core gamma units capable of providing total and spectral logs



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Roller Gamma Logger



The GMS312 Core Analyser is a rolling Gamma Spectrometer designed to detect naturally occurring gamma radiation emitted by core samples.

The Instrument is used to monitor gamma radiation over a user defined period and display the results in four predefined Regions of Interest (ROI). The predefined ROI are Uranium, Potassium, Thorium and a total gamma (total spectrum counts). It is also possible to add user defined ROI in 3 additional channels available on the instrument.

Survey results are saved manually or automatically to the internal memory of the instrument, so that they may be downloaded and analysed on a PC via the GMS Core Analyser Data Manager Application software.

The GMS312 is intended as a first look tool for analysis of a core. It is not intended to be used a laboratory standard instrument and does not provide definitive results. The Regions of Interest provide data that should give an experience core analyst the means to correlate the GMS312 data with wire line measurements.

ROI Window	Name	Description	Lower Limit (keV)	Upper Limit (keV
1	Uranium	Uranium Peak (1760keV)	1596keV	2024keV
2	Potassium	Thorium Peak (580keV)	493keV	667keV
3	Thorium	Potassium Peak (1460keV)	1241keV	1596keV
4	(Blank)	(Blank)	(Blank)	(Blank)
5	(Blank)	(Blank)	(Blank)	(Blank)
6	(Blank)	(Blank)	(Blank)	(Blank)
7	Spectrum	Spectrum Total	90keV	2100keV

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CORING ACTIVITIES
AT RIG

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Preparation – Pick Up Coring BHA

Initial Process :

- Coring Personnel To Join All Scheduled Rig Meetings after arrival
- Working Permit – Open by Coring Personnel
- Coring JSA to be distributed to All related at Rigsite, by Coring Personnel
- Tool Box Meeting before activities



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Make Up Core Barrel with Durateq



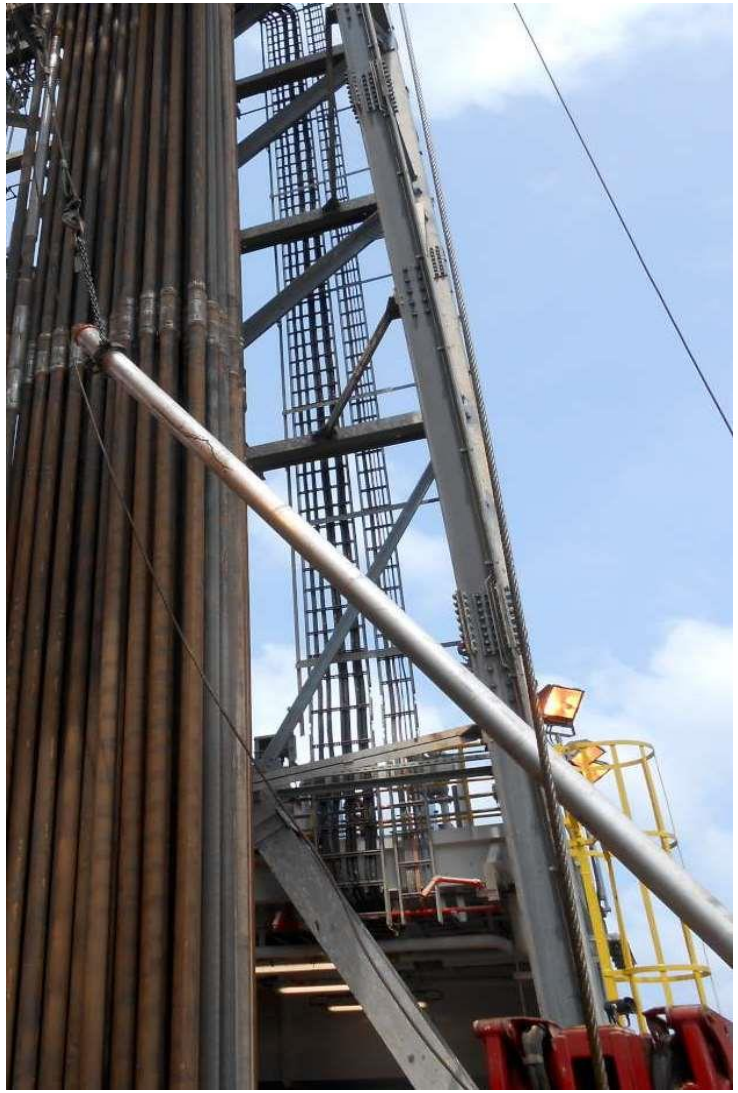
Pick up Core Barrel



Make Up Torque Core Barrel and Stabilizers

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Make Up Inner Barrel to LAS with Durateq



Pick Up Inner Barrel



Attached Clamp after sliding Inner Barrel inside Outer Barrel



Hand Tight Torque

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Spacing Inner Barrel for Durateq



Spacing Inner Barrel, If the groove of Split Ring not aligned

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Make Up Core Head to Core Barrel with Durateq



Slowly aligned assembly to Core Head



Make Up Torque Core Head to the Lower Sub and Upper Sub



Double Check Core Head Condition and Duraclose Shoe Position after MUT.

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Make Up Core Barrel with Duraclose



Tilt and Attach Top Drive Elevator to Core Barrel's Elevator Sub



Make Up Torque Core Barrel and Stabilizers



Make Up Upper Sub for Duraclose, then Run In Hole Shortly

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DAS (Duraclose Activation System to the Inner Barrel)





Disconnect DAS from Outer Barrel Assy.

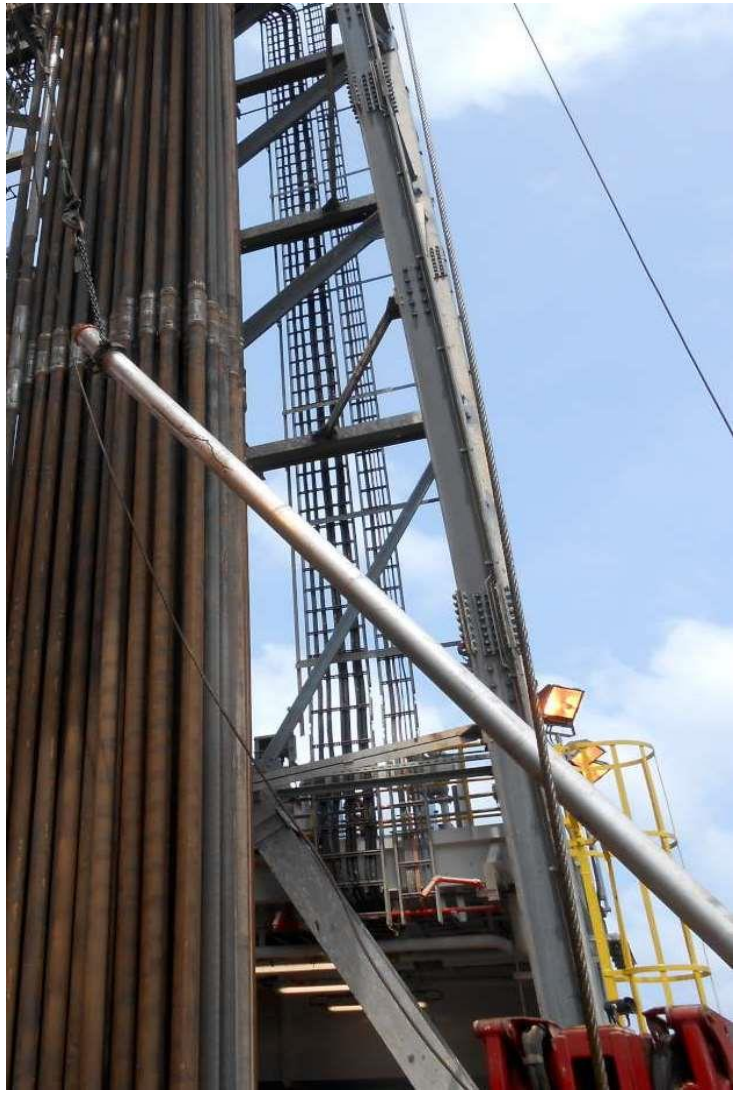
Double Check Bypass Valve position is still CLOSING the Port (No Port is exposed),

before raise up DAS assembly Pin) to connect to Inner Barrel (Box), later on

Disassemble DAS from Core Barrel

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Make Up Inner Barrel to DAS



Pick Up Inner Barrel



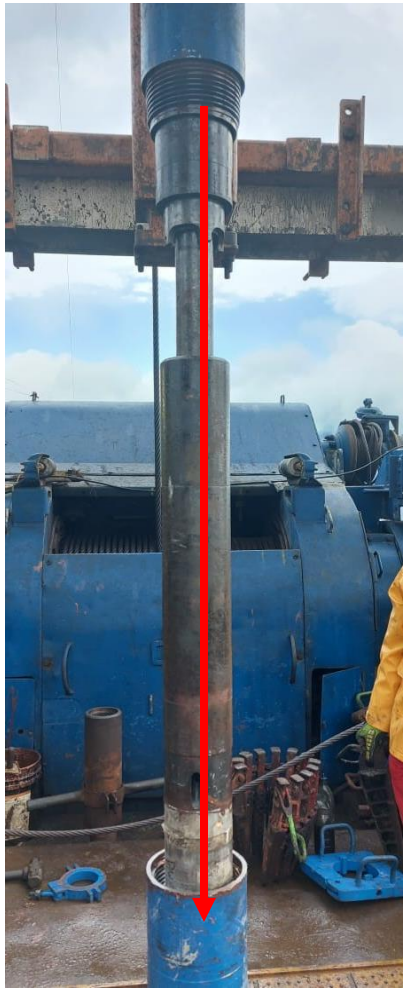
Attached IT Clamp, after sliding Inner Barrel inside Outer Barrel



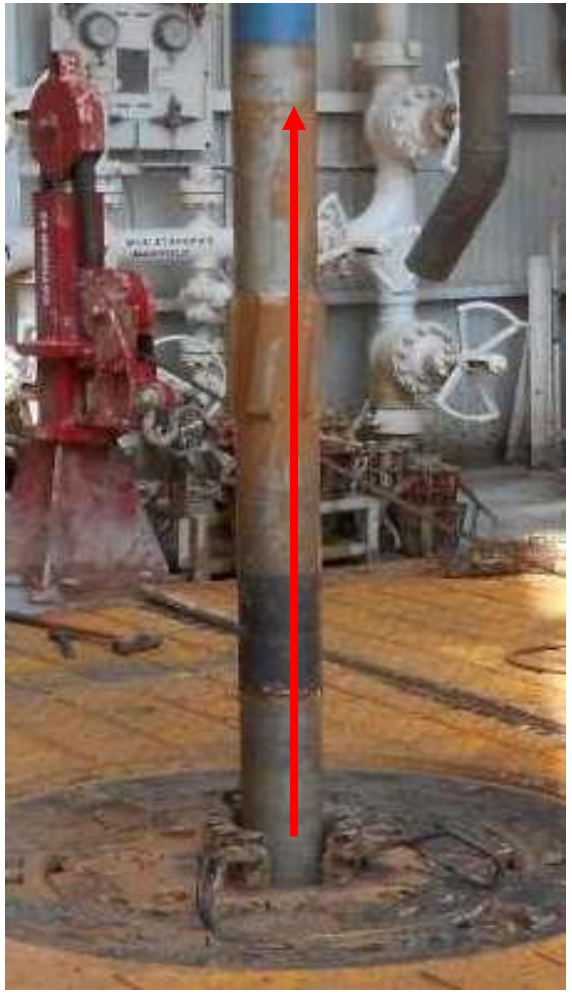
Hand Tight Torque

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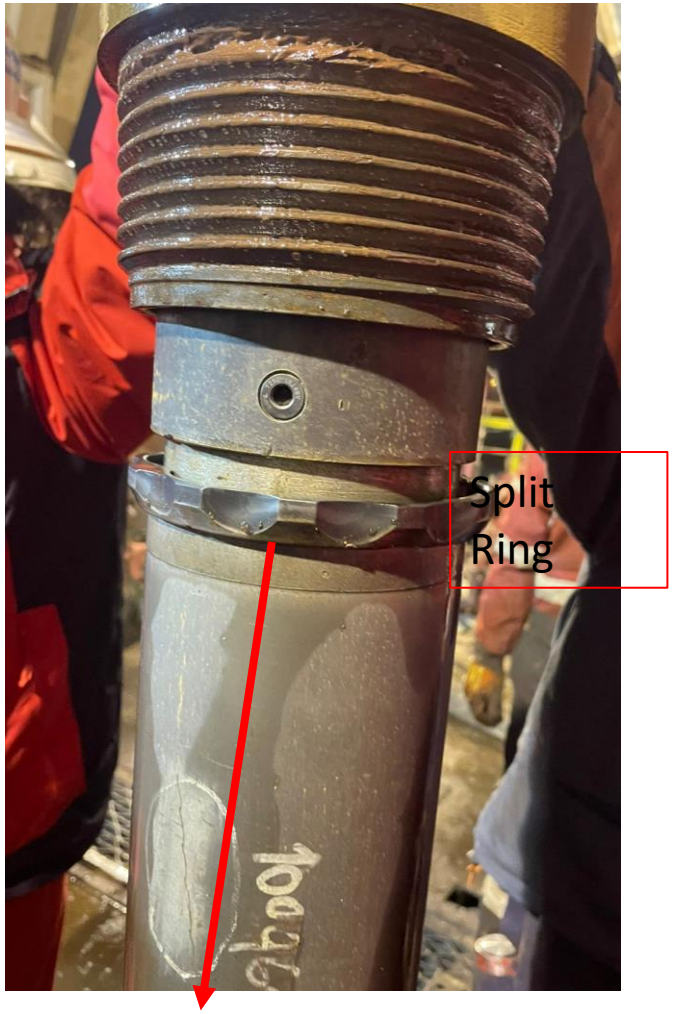
Make Up Split Ring at Duraclose Shoe



Run In – DAS & Inner Barrel Assy Down to Outer Barrel Assy, Make Up Torque DAS to Barrel



Coring BHA pick up from Hole again, until DCS is present,



Install Split Ring, between groove of Upper Sub and Lower Sub, as part of Activation system for Duraclose Assembly.



Spacing Inner Barrel necessary, with LAS Wrench, If the Split Ring, unable to fit into the groove

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Make Up Core Head to Core Barrel



After Make Up Torque Lower Sub, Slowly align assembly to Core Head



Make Up Torque Core Head to the Lower Sub and Upper Sub



Double Check Core Head Condition and Duraclose Shoe Position after MUT.

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R.I.H

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Identified Risk and Mitigation

No.	Risk	Mitigation
1	Unconsolidated Formation	Full Closure System - Duraclose
2	Swelling Reactive Shale / Sticky Clay that could induce Core Jamming	Mud System – SOBM / Past nearby Coring Job Parameter as reference
3	Washed Out Core / Milling after core jamming	Low Parameter / Mitigate Jamming Itself / Past nearby coring job parameter as reference
4	Junk in Hole while Coring	Cleaning Trip before RIH BHA Coring

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Coring Operation with Durateq – for Consolidated

- Bottom Hole parameter at Coring Point, information will be double checked onsite by Coring personnel.
- Pre-Hole Coring must have **no debris/junk** and follow recommendation within Coring Program
- With HPWBM, Recommend to maintain Water Loss <4cc/30min. Highly Recommended to use SOBM or OBM, while Coring to further help minimize Core Jamming
- Coring parameters and procedures will recommend for fractured,
  - Make-up Core Barrel c/w **DURATEQ System**, and RIH
  - Tag Bottom,Check Off Bottom parameter, then Drop **1st Coring Ball** (1-1/4" OD) with 150GPM to divert mud flow
  - Inform WSGeologist, Company-man, Driller prior commencing Start of Coring.

    @Coring Parameter Range :

    Flow Rate : 200 - 300 GPM

    WOB : 4 – 30 klbs

    RPM : 70 - 100
  - After TD coring 30ft, Stop Coring, Drill Off until Zero WOB, Stop Rotary. Stop Pump, Break Core P/U BHA Off Bottom
  - POOH slowly with timed **as per Tripping Out time** schedule. (Could activate Circulating Sub upon necessary, prior LCM)
  - Retrieve Core inside Inner Barrel, Lay down at yard, Measure Core Recovery
  - Mark Top and Bottom Depth for each footage at the inner barrel.
  - Cut Inner Barrel into 3ft length
  - Preservation by Foam Resination
  - Packaging the cutted 3ft length inner barrel into Wooden Core Boxes

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CONVENTIONAL CORING TOOLS “DURATEQ” VIDEO



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Coring Operation with Duraclose for Unconsolidated

- Bottom Hole parameter at Coring Point (Preferred Sandstone), information will be double checked onsite by Coring personnel.
- Pre-Hole Coring must have no debris/junk and follow recommendation within Coring Program
- With HPWBM, Recommend to maintain Water Loss <4cc/30min. Highly Recommended to use SOBM or OBM
- Coring parameters and procedures will recommend for unconsolidated formations,
  - Make-up Core Barrel c/w DURACLOSE System, and RIH
  - Tag Bottom,Check Off-Bottom parameter, then Drop 1st Coring Ball (1" OD) with 150GPM to divert mud flow

    @Coring Parameter :

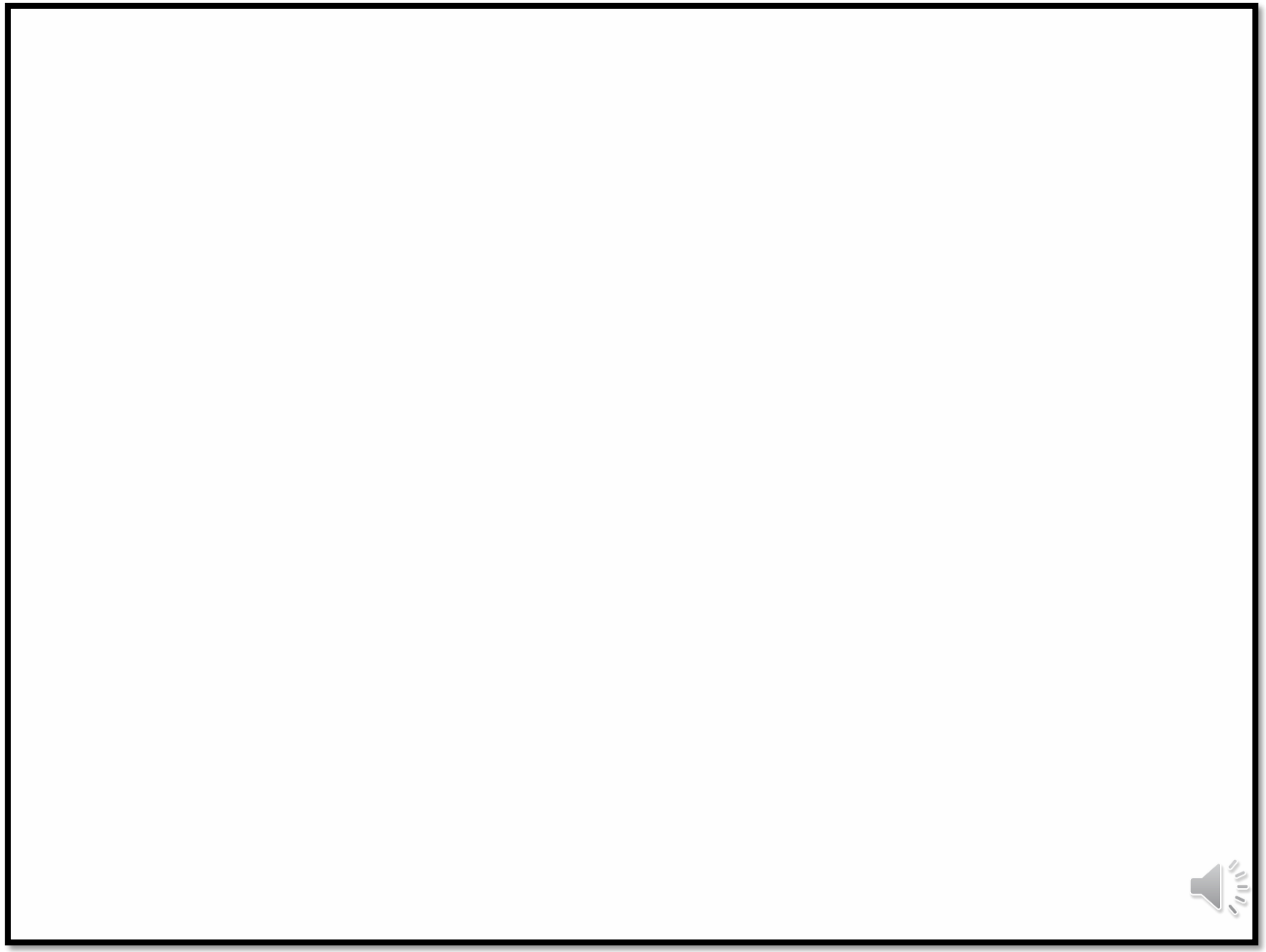
    Flow Rate : 150 - 200 GPM

    WOB : 4 – 20 klbs

    RPM : 70 - 100
  - After TD coring 30ft, Stop Coring, Drill Off until Zero WOB, Stop Rotary, **Stay On-bottom**.
  - **Drop 2nd Ball** (1-1/4" OD) with 150-200 GPM of Flow rate, after pressure peak seen, wait few minutes to make sure Duraclose is activated, by stable SPP.
  - POOH slowly with timed Tripping Out time schedule.
  - Retrieve Core along-side the Inner Barrel, Lay down at deck, Measure Core Recovery
  - Mark Top and Bottom Depth for each footage at the inner barrel.
  - Cut Inner Barrel into 3ft length section
  - Preservation with Resination (Injection method)
  - Packaging the cutted 3ft length inner barrel into Wooden Core Boxes. Mark the Wooden Boxes

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CONVENTIONAL CORING TOOLS “DURACLOSE” VIDEO



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Required from Rigsite

- Stable 1 ea Mud Pump prior Coring , need support to double check on Mud Pump efficiency before and while coring.
- With Durateq – If Target formation is consolidated formation, need capability of low flow rate at least of 200 GPM if not lower once required from Mudpump.
- With Duraclose - If Target formation is confirmed Unconsolidated formations, need capability of low flow rate at least of 150 GPM if not lower once required (Need to set up Mud Pump or Bypass the flow rate).
- Pup Joint available once required to keep the Kelly Up prior Coring needs to be a minimum 9m – 18m, without stopping halfway that creates any connections in between while coring in progress.
- All surface gage are calibrated, especially on SPP and TRQ
- Driller to support Coring Engineer recommendations, to apply certain coring parameters fluently, to prevent any drill-off while coring.
- Driller to support the Coring BHA's Trip Out time schedule as per program to be followed.
- Any Core Jamming anomaly that is confirmed by the parameters, Coring Engineer would discuss with all related (Geologist, Company-Man, Driller). Stop Coring and POOH soonest is recommended, to prevent any escalation to the potential reduced core recovery.
- Rig Crew to not use or apply any downhole rotation while connection in progress at POOH activity. Kelly Spinner is recommended.
- Core Processing Area near Air Compressor supply, along-with 13m (Length) x 4 m (Width) area, is required. A minimum of 80 psi is required to run the Air Saw machine with good performance.

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P.O.O.H

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H2S Mitigation



Sniff H2S, with H2S detector, by a H2S Contractor with SCBA (Other 3rd party). – **From the Last 10 Stands** prior POOH Core Barrel Assembly



Blower ON



Use of SCBA for Coring Crew once/if necessary (H2S Rep to inform), prior Disassembling Core barrel

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Core Barrel POOH Reach Surface with Duraclose





Install Elevator Sub to Barrel, Slowly POOH Coring BHA until above Rig Floor, To check Core inside, and BHA condition



Disconnect Core Head and Lower Sub

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Surface Handling with Duraclose





Pick Up BHA Coring and REMOVE SPLIT RING Before process of Pulling Out Inner Barrel from Core Barrel.



Slowly Run Coring BHA Back In Hole, until at DAS position, Attach Drilling Slip and Dog Clamp, Assemble Elevator Sub to Core Barrel, Disconnect DAS (Duraclose Activation System) from Top Stabilizer.

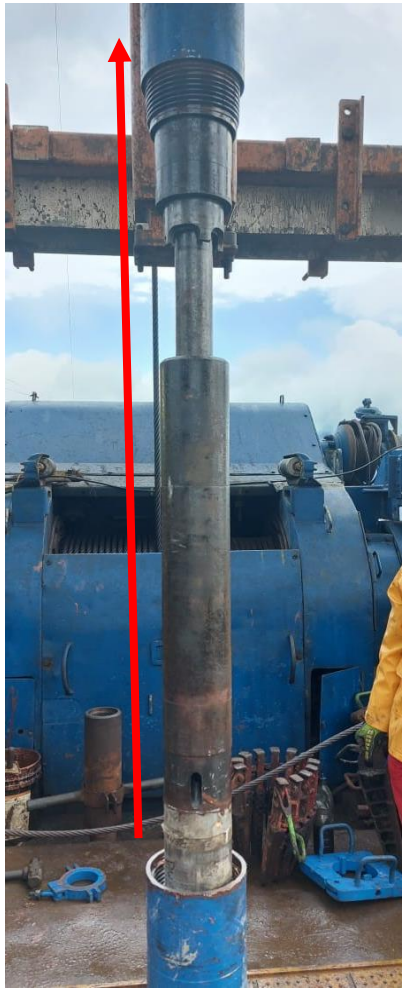
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Duraclose Activation System (DAS)



DAS Port – At Open Position

Means Duraclose System Activated Properly



Raise Up DAS with Inner Barrel Assy

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Surface Handling



Attached Inner Tube Clamp
Disconnect DAS vs Inner Barrel





Install Lifting Nubbin to Inner Barrel Box Conn.) – Then Pick Up



POOH Inner Barrel from Core Barrel

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Surface Handling





Non Man Riding Laydown Cradle

- Eliminate bending movement of barrel
- Removes Man Riding requirement
- Hydraulically activated “fingers” hold Inner barrel in place.





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Core Sample Processing

Measuring and Marking, of Core Sample

- Will be conducted by Geologists Team on board.
- Before cutting process, Core must be handled & prepared properly, together with Coring Engineer



REQUIRED:

- Geologist Team and Experienced Field Technicians
- Order of process – clean, measure and mark.
- Sufficient equipment to continue further process properly.

- If H2S still present, Isolate Core Processing Area with safety Line, with Blowers On, and wait until No H2S hazard detected by H2S detector (H2S Representative required).

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RESIN LIQUID - Preservation



Uses 2 (Two) Chemicals – Combined Equally
Past Cast A149 (Resin) and Past Cast B149 (Hardener)



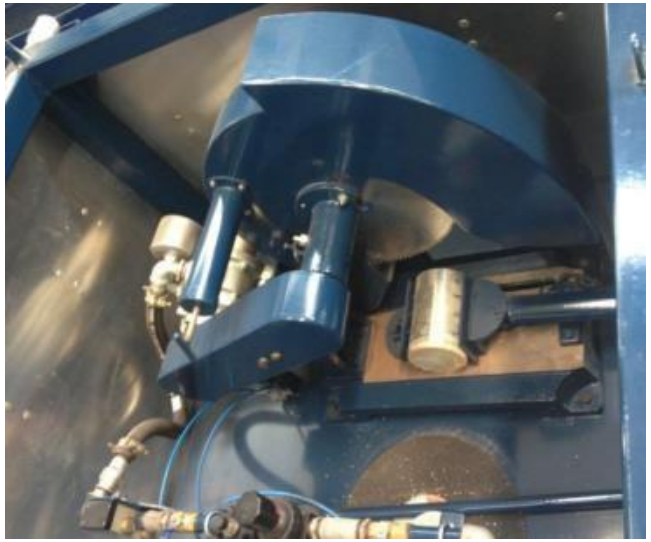
Resin both ends, and capped both ends

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Core Sample Processing

Air Saw – Cutting Process

- Box style with Diamond & Abrasive Blades









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Packaging – Wooden Core Boxes

Core Crates

- Profile Cut Foam Padding
- 1m per wooden box



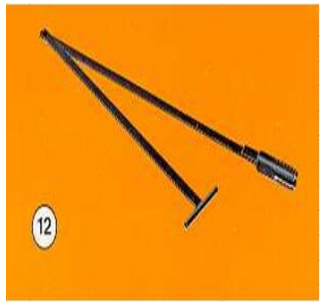
CORE STORAGE
AT LAB STORAGE
AREA



AT LAB FOR STORAGE

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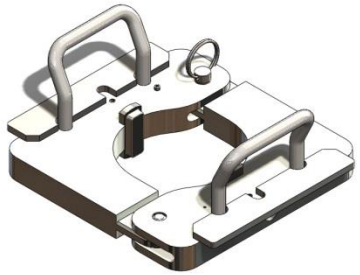
Handling Tools



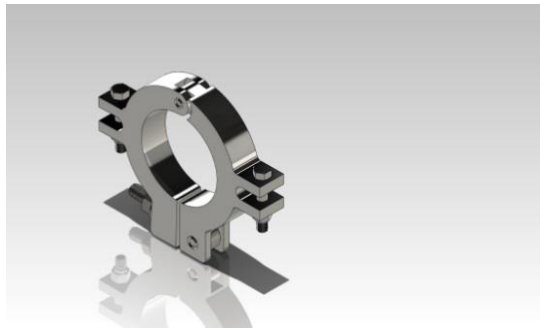
Drop Ball Retriever



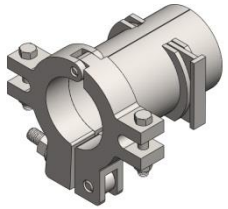
Pneumatic Air Saw



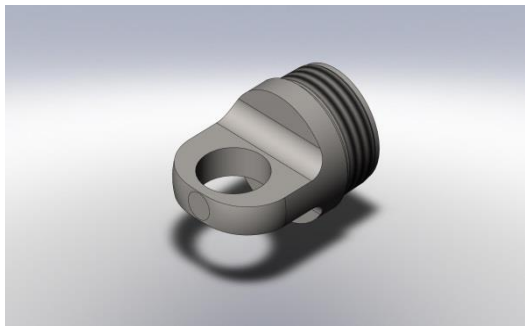
Bit Breaker



Inner Tube Clamp



Guillotine Boot



Lifting Bail / Plug



Electric Saw



LAS Wrench

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Knowledge Check

DOWNHOLE
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